Quantum theory of actively-phase-locked optical parametric oscillators subject to limit-cycle motion

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A thesis submitted to the Friedrich-Alexander-University in accordance with the requirements of the degree of MASTER OF SCIENCE.

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SEPTEMBER 2018

Abstract

Quantum mechanics holds the promise of a second technological revolution, one driven by a special type of quantum correlations not present in the classical framework, denoted by *entanglement*. Optical parametric oscillators are a type of nonlinear optical devices that constitute the most reliable source of entangled light. As such, they are of great important for the fields of quantum information, quantum optics, and quantum technologies. This thesis focuses on the quantum theoretical description of a special type of optical parametric oscillators, dubbed *actively-phase-locked*, which have the distinctive property of undergoing limit-cycle motion in a certain parameter regime. In regions where they reach stationary behavior, the quantum properties of these devices were studied through the usual linearization technique, which assumes that quantum fluctuations around the classical states are small, a generally good approximation for nonlinear optical cavities. In contrast, the nontrivial periodic time dynamics associated to limit-cycle motion offers a big challenge, because the fluctuations cannot be considered small along the closed trajectory drawn by the cycle in phase space. In this thesis we apply a recently developed technique that extends the linearized approach to such situations. The technique allows us to approximate the asymptotic quantum state by a mixture of Gaussian states centered at the points of the cycle's trajectory. Analyzing in detail the quantum correlations of these Gaussian states, we show that actively-phase-locked optical parametric oscillators produce light within the cavity with large entanglement levels, even in the regions where they undergo limit cycle motion. The results found in this thesis open then the way for an experimental observation of entanglement in such a dynamically nontrivial scenario.

Dedication and acknowledgements

I would like to thank my father for instilling a thirst for reasoning, through numerous debates and discussions. I would like to thank my mother for unfailing support and continuous encouragement throughout my years of study.

I would also like to thank my thesis advisor Dr.Carlos Navarrete-Benlloch for the opportunity to work on such an interesting project. The door to Carlos's office was always open both literally and figuratively (through whatsapp) whenever I ran into trouble spot or had any question.

Last but not the least I would like to thank the office of graduate school in advanced optical technology especially Dr. Jürgen Großmann for helping me through the bureaucratic processes at every step.

Author's declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Research Degree Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, the work is the candidate's own work. Work done in collaboration with, or with the assistance of, others, is indicated as such. Any views expressed in the dissertation are those of the author.

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CHAPTER 1

INTRODUCTION

Nature strives to minimize variations of a certain quantity known as the action. Up until the end of the XIX century, systems were thought to move along well-defined paths with minimum variance in this quantity, an assumption that served as the basis for most of the formalism of classical mechanics. These foundations were confronted at the advent of the XX century with the reality put forward by some experiments, which could be explained only by allowing for paths other than the one minimizing the action to take an active role in the system dynamics. This culminated with the formulation of quantum mechanics by the end of the first quarter of the century. These framework suggested that the microscopic world is ruled by fascinating laws that allow systems to present themselves both as waves (when evolving unperturbed) and as particles (when observed through a measurement). The wave nature was most clearly put forward by Schrodinger in his famous equation [1], the solution of which is a complex wave function that presents interference phenomena, and whose absolute value square was interpreted by Born as the probability distribution of the paths that the quantum system has access to.

In 1935 Einstein, Podolsky, and Rosen (EPR) put out a paper [2] that was set to prove the incompleteness of quantum mechanics, and of the information provided by the wave function in particular. To this aim, they introduced a very special quantum state of a bipartite mechanical system, which shows perfect correlations between both the positions and momenta of the two subsystems, say A and B. Then they argued as follows: if the position is measured in A, then B collapses to a well defined position eigenstate. If, instead, the momentum of A is measured, then B collapses to a well-defined momentum eigenstate. According to the postulates of quantum mechanics, this is true even if subsystems A and B are spatially separated. Hence, they proceed, if we accept that no action effected on

A can affect B (no *spooky action at a distance*), then we must conclude that the position and momentum of B where well defined all along. However, this is in contradiction with the uncertainty principle that follows also from the postulates of quantum mechanics, and therefore we seem to be forced to conclude that the description provided by the wave function is incomplete.

At the time, the EPR arguments went relatively unnoticed, except for a few hot debates between Einstein, Bohr and a few supporters of their disparate views. It had to pass 30 years until John Bell brought the EPR paper to the spotlight, but refining it in a brilliant way [3]: in a nutshell, he proved that no 'complete' description which includes locality (what we now refer to as *local hidden-variables theories*) can reproduce the predictions of quantum mechanics. In other words, any "more complete" theory of the world must necessarily come with *non-local* effects, which feels even more troubling than the "incompleteness" or probabilistic nature of quantum mechanics. Bell even provided a set of inequalities obeyed by any local hidden-variables theory, but which can be violated by quantum mechanics, opening the door for an experimental discrimination between such alternatives. Of course, such experiments have always ruled out in favour of quantum mechanics, first in setups containing loopholes [4, 5], but more recently in a series of loophole-free tests [6–8] that have settled the question for good according to most people.

In spite of their wrong conclusion, the EPR paper has been one of the most influential papers for the quantum-mechanical ideas developed over the last few decades, specifically in the relatively young field of quantum information and its implementations based on quantum-optical and atomic systems. This is because the bipartite state they came up with (now known as *EPR state*) was the first one to show a new type of correlations beyond what's attainable in classical physics. Nowadays, we call this type of correlations *entanglement*, which fuel the most promising future technological applications of quantum information to computing, simulation, sensing, and communication.

Nonlinear optical processes are one of the main source for generating quantum states of light, as they open up a realm of physics with effective photon-photon interactions. Specifically, the generation of entangled states of light relies on the nonlinear process known as parametric down-conversion, which occurs in birefringent crystals with second-order nonlinearity. Unfortunately, the nonlinearities of such systems are weak in general, so in order to increase them effectively, the nonlinear crystal is placed inside an optical cavity. The resulting systems are known as optical parametric oscillators (OPOs), which are the main subject of study in this thesis. Specifically, we deal with type II OPOs, which are capable of generating two entangled optical beams with orthogonal polarization and different frequencies in general. In [9] it was shown that using an additional laser to drive the cavity, the frequencies of the entangled beams can be made equal, which is highly desirable for experiments. Such devices were dubbed *actively-phase-locked* OPOs. It was shown that entanglement is preserved to a high degree in a large region of the parameter space. However, in [9] it was possible to study the quantum properties of this system only when it reaches a stationary state in the classical limit. But in the same reference, it was shown that there is a large region of the parameter space where the system shows self-pulsing behavior, leading to limit-cycle motion. The purpose of this thesis is to fill this gap, and study the quantum properties of actively-phase-locked OPOs in the regions where they undergo limit-cycle motion in the classical limit.

The usual method used to analyze quantum properties of nonlinear optical cavities is called *linearization*, in which one assumes that quantum noise induces "small" fluctuations around the classical steady state. The method then provides a Gaussian state centered around the classical solution, which approximates well the true quantum state of the system, except in very specific points of the space of parameters (such as large optical nonlinearities or bifurcations, which we will introduce later). Unfortunately, the standard linearization technique is not applicable around limit-cycle motion, because fluctuations cannot be considered small along the cycles trajectory, as we will show explicitly along the thesis. For this situation, a generalized linearization technique was introduced in [10], which was applied in the same reference to a paradigmatic nonlinear model, the quantum Van der Pol oscillator. This model, however, is not directly connected to current experimental implementations. Therefore, here we apply this linearization technique for the first time to a model with direct connection to experiments. Moreover, while the quantum states obtained for the Van der Pol oscillator are relatively trivial, we will show that in the case of OPOs, we obtain states with large levels of entanglement.

The thesis is organized as follows:

- **Chapter 2.** We start describing in detail the nonlinear optical system under consideration. In particular, we introduce the basic Hamiltonian that governs the evolution of the actively-phase-locked OPO, seen as a cavity interacting with external quantum fields, and then derive a master equation for the state of the cavity alone seen as an open system.

- **Chapter 3.** We then introduce the concept of quasiprobability distributions, which allow us to interpret in phase space the quantum dynamics of the system. We show how to map the master equation to a partial differential equation that governs the evolution of such distributions.

- **Chapter 4.** It is then shown that the latter equation can be mapped to a set of stochastic Langevin equations by using a generalized phase-space representation known as the positive P distribution. For this purpose, in the first part of the chapter we introduce Fokker-Planck equations and their connection to stochastic Langevin equations. After that, we introduce the positive P representation generalizing the phase-space distributions of the previous chapter, and show that it satisfies an evolution equation that has the form of a Fokker-Planck equation.

- **Chapter 5.** As a first, necessary step, in this part of the thesis we discuss the phase diagram of the actively-phase-locked OPO in the classical limit, showing the regions where limit-cycle motion occurs.
- **Chapter 6.** With the stochastic Langevin equations and the solutions in the classical limit at hand, in this chapter we apply the linearization technique introduced in [10] to the limit-cycles appearing in the actively-phase-locked OPO. In particular, we show how this technique allows us to obtain an approximate quantum state in the form of a balanced mixture of Gaussian states defined at the points of the closed trajectory that the limit-cycle describes in phase space.
- **Chapter 7.** In this chapter we analyze in detail the Gaussian states that form the mixture presented in the previous chapter. Specifically, we analyze their quantum correlations and comment on the possibility of observing or using the entanglement that we predict.
- **Chapter 8.** Finally, we provide some conclusions and comment on the ideas that will be pursued in the future and naturally follow from the results of this thesis.

CHAPTER 2

MODELLING OUR SYSTEM

The system analyzed in this thesis is a so-called actively phase-locked type II optical parametric oscillator. In this chapter we introduce it in detail, and justify the basic equations that we will use to model it both quantum mechanically and classically.

2.1 Nonlinear optics

Nonlinear processes occur when light interacts with a medium which responds to it a nonlinear manner, i.e., the induced polarization of the medium depends nonlinearly on the light's electric field $E(t)$ (we stay in the scalar approximation to simplify the notation for now) [11]:

$$P(\mathbf{r},t) = \underbrace{\epsilon_0\chi^{(1)}E(\mathbf{r},t)}_{P_1(\mathbf{r},t)} + \underbrace{\epsilon_0\chi^{(2)}E^2(\mathbf{r},t)}_{P_2(\mathbf{r},t)} + \underbrace{\epsilon_0\chi^{(3)}E^3(\mathbf{r},t)}_{P_3(\mathbf{r},t)} + ... \tag{2.1}$$

Here, ϵ_0 is the permittivity of free space and $\chi^{(n)}$ is the n-th order susceptibility, the linear one $\chi^{(1)}$ being the most common light-matter interaction, responsible for the refractive index given by $n^2 = 1 + \chi^{(1)}$. The nonlinear terms of the above expression generate a polarization that has frequencies at different combinations of the frequencies of the light that interacts with the medium. For example, let us assume the light has two frequencies ω_p and $\omega_s < \omega_p$, the polarization of the medium due to second order interaction, $P_2(\mathbf{r},t)$, will contain frequencies such as $2\omega_p$, $2\omega_s$, $\omega_p + \omega_s$ and $\omega_p - \omega_s$, which generate light at the corresponding frequencies, as seen from Maxwell's macroscopic equations [11, 12]:

$$\nabla^2 E - \frac{n^2}{c^2}\frac{\partial^2 E}{\partial t^2} = \frac{1}{\epsilon_0 c^2}\frac{\partial^2(P_2 + P_3 + ...)}{\partial t^2} \tag{2.2}$$

We can already draw some intuition about why nonlinear processes can be used to generate quantum states: energy conservation implies that the photons at the new frequencies are generated by "dividing" or "combining" photons of the original frequencies via nonlinear processes. The nonlinear process which generates frequency $\omega_p - \omega_s$ is the most common tool for the creation of quantum states, and is known as parametric down-conversion (PDC). The medium is typically pumped with a strong field at the higher frequency ω_p (denoted by "pump"), while a weak field is injected at the lower frequency ω_s (denoted by "signal"). This induces the generation of the so-called "idler" frequency $\omega_i = \omega_p - \omega_s$. If no light at frequency ω_s is injected, then we talk about "spontaneous" parametric down-conversion, and the distinction between signal and idler is arbitrary. The frequencies of signal and idler are selected by the so-called phase-matching condition, which can also be thought of as momentum conservation for the nonlinear process. This relation can be found by substituting $P_2(\mathbf{r}, t)$ in (2.2):

$$\mathbf{k}_p = \mathbf{k}_s + \mathbf{k}_i \tag{2.3}$$

where $\mathbf{k}_j$ is the wave vector of the corresponding field component. Deviations around this condition must be small in order to achieve high conversion efficiency. Attaining this condition for a desired combination of frequencies is made especially difficult by the dispersion of the medium. Nevertheless, it is possible to do so by using a birefringent crystal, possessing different refractive indices n_o and n_e along two orthogonal axes called "ordinary" and "extraordinary". The phase-matching condition reads then as

$$n(\theta)\omega_p = n_e\omega_s + n_i\omega_i \tag{2.4}$$

where it assumed that the signal is polarized along the extraordinary axis, while the idler can be polarized along either the extraordinary ($n_i = n_e$) or ordinary ($n_i = n_o$) axis. The pump is linearly polarized forming an angle θ with respect the extaordinary axis, which is chosen such that the phase-matching condition is satisfied for the desired combination of frequencies. When signal and idler have the same or different polarization, we talk, respectively, about type I or II phase matching. The latter will be the case that we will focus on in this thesis.

2.2 Generation of quantum states from nonlinear optics

In this thesis we will study type II parametric down-conversion quantum mechanically. As sketched above, at the most fundamental level this process can be understood as a pump photon being devided into a signal-idler photon pair. When the pump field is treated classically, the Hamiltonian for this process reads then as [11, 13]

$$\hat{H} = \mathrm{i}\hbar G\hat{a}_s^\dagger\hat{a}_i^\dagger + \mathrm{H.c.}, \tag{2.5}$$

where G is a complex parameter with a phase inherited from the pump's phase and an absolute value proportional to the nonlinear susceptibility $\chi^{(2)}$, the crystal's length,

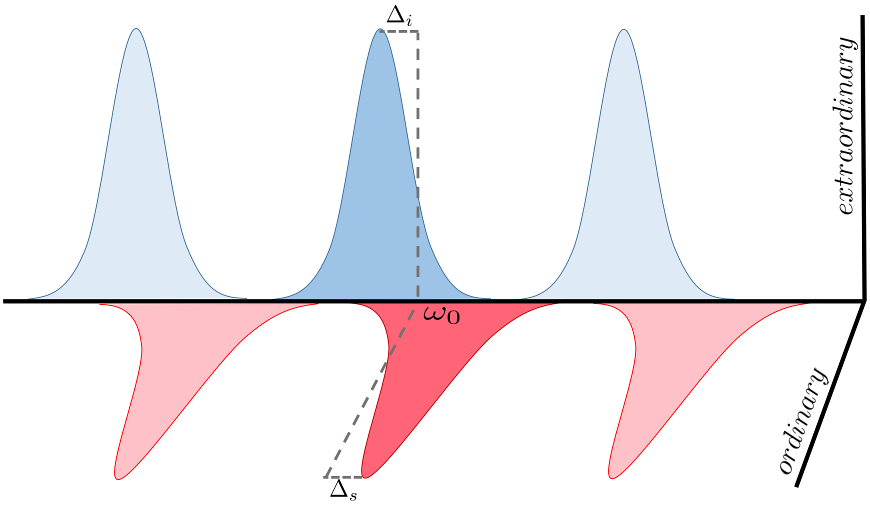

FIGURE 2.1. Resonance scheme of the type II OPO considered in this work. The birefringence of the crystal breaks the degeneracy between the modes with ordinary and extraordinary polarization, which experience different refractive indices. The mirrors of the cavity are transparent to the pump of frequency $2\omega_0$, and we also drive the cavity with a laser at frequency ω_0 (active locking configuration).

and the square root of the pump power. The time evolution operator associated to the above Hamiltonian corresponds to the so-called two-mode squeezing operator with a time-dependent squeezing parameter Gt. We will study this operator and its physical consequences in detail in Chapter 7. For now, it is interesting to note that quantum correlations (entanglement) between signal and idler can be induced by applying it to their quantum state.

However, since in general the nonlinear interaction is weak, it is common to place the nonlinear crystal inside a cavity (a configuration commonly known as "optical parametric oscillator" or OPO), such that the light-matter interaction is effectively enhanced. In turn, this modifies the simple Hamiltonian exposed above, introducing terms accounting for the interaction with the field outside the cavity, which ultimately induce dissipation and make the time evolution of the density operator non-unitary. We will introduce this formalism in detail along the next sections. Interestingly, apart from increasing the nonlinear interaction and inducing dissipation, the cavity introduces a threshold condition in the form of a pitchfork bifurcation. For a pump power below this threshold, the cavity remains in a two-mode squeezed vacuum state, while above threshold it gains a coherent contribution. At threshold and for an idealized system, the light coming out of the cavity can achieve perfect squeezing. A record squeezing of 15dB (97% quadrature noise reduction) was achieved by a type I OPO with a semi-monolithic linear cavity formed by an external coupling mirror and the back surface of a 9.3 mm long periodically-poled

KTP crystal [14].

2.3 Actively-phase-locked optical parametric oscillators

The system considered for this project is a type II OPO under active phase-locking conditions [9], meaning that both signal and idler receive external injection, aimed at locking their frequencies to the degenerate one, $\omega_s = \omega_i = \omega_p/2$, as we explain later. In order to simplify the analysis, and without loss of generality, the mirrors of the optical cavity are assumed transparent at the pump frequency.

Some issues appear in frequency non-degenerate OPOs with the detection of squeezing in a balanced homodyne detection scheme. First, because the relative phase between signal and idler can diffuse rather fast. Second, because when signal and idler have different frequencies, it is required a local oscillator with those exact frequency components.

There have been successful attempts to obtain frequency degeneracy and phase locking by coupling signal and idler via a $\lambda/4$ wave plate embedded in the cavity [15–17]. This type of OPO is called self-phase-locked OPO. In this project we will be considering a different, non-invasive frequency locking technique consisting on the injection of a laser field at half the pump frequency. This type of OPO is called actively-phase-locked OPO [9].

Let us now pass to describe in detail the model we use for this system. First, we consider only the parts of the electric field operator of relevance for our system, writing it as a sum of three terms (note that we implicitly assume to work in the Schrödinger picture, where operators do not evolve):

$$\hat{\mathbf{E}}(\mathbf{r}) = \hat{\mathbf{E}}_c(\mathbf{r}) + \hat{\mathbf{E}}_b(\mathbf{r}) + \hat{\mathbf{E}}_p(\mathbf{r}). \tag{2.6}$$

$\hat{\mathbf{E}}_c(\mathbf{r})$ is the electric field inside the cavity, while $\hat{\mathbf{E}}_b(\mathbf{r})$ and $\hat{\mathbf{E}}_p(\mathbf{r})$ are the electric fields outside the cavity at the frequencies surrounding that of the down-converted modes and the pump laser. Let us now write these terms in detail.

As explained, we work under type II conditions, and therefore consider two cavity modes with orthogonal polarization. Further, we assume that the nonlinear crystal is thin compared with the cavity length, and it's placed at the cavity waist. For longitudinal coordinates z close to the center of the crystal (defining $z = 0$), the cavity field is then written as the sum of two terms $\hat{\mathbf{E}}_c(\mathbf{r}) = \hat{\mathbf{E}}_{cs}(\mathbf{r}) + \hat{\mathbf{E}}_{ci}(\mathbf{r})$, where [18]

$$\hat{\mathbf{E}}_{cj} = \mathrm{i}\sqrt{\frac{\hbar\omega_j}{\epsilon_0 L n_j}}\,\boldsymbol{\epsilon}_j A(x,y)\hat{a}_j e^{\mathrm{i}k_j z} + \text{H.c.}. \tag{2.7}$$

In this expression, L is the length of the cavity, n_j is the refractive index experienced by the corresponding mode, $\boldsymbol{\epsilon}_j$ being its polarization. $\hat{a}_j$ and $\hat{a}_j^\dagger$ are the corresponding annihilation and creation operators, which satisfy canonical commutation relations $[\hat{a}_j, \hat{a}_l^\dagger] = \delta_{jl}$. $A(x,y)$

is the transverse profile of the electric field (assumed the same for signal and idler for simplicity), which we normalize as $\int dxdy|A(x,y)|^2 = 1$.

Let us move on to the field outside the cavity. In principle, there is a continuum of modes extending over all possible frequencies, transverse profiles, and polarizations. However, we only care about the modes with the same polarization and transverse profile as signal and idler, and frequencies around their resonance frequency in the cavity, as only these modes will be excited when photons leak out through the partially transmitting mirror. The corresponding electric field can be written as $\hat{\mathbf{E}}_b = \hat{\mathbf{E}}_{bs} + \hat{\mathbf{E}}_{bi}$, with [18]

$$\hat{\mathbf{E}}_{bj} = \int_{\mathcal{O}(\omega)} d\omega \, \mathrm{i}\sqrt{\frac{\hbar\omega}{\epsilon_0 c\pi}} \boldsymbol{\epsilon}_j A_{bj}(\mathbf{r})\hat{b}_j(\omega)e^{\mathrm{i}kz} + \text{H.c.}, \tag{2.8}$$

where $\mathcal{O}(\omega_j)$ indicates the integration over the frequency in the vicinity of ω_j, and $A_{bj}(\mathbf{r})$ are the transverse modes that match the profile of the cavity modes at the exit mirror, as well as appropriate boundary conditions. $\hat{b}_j(\omega)$ and $\hat{b}_j^\dagger(\omega)$ are the corresponding annihilation and creation operators which satisfy continuous canonical commutation relations $[\hat{b}_j(\omega), \hat{b}_l^\dagger(\omega')] = \delta_{jl}\delta(\omega - \omega')$.

Finally, we consider the external modes for frequencies around the one of the pumping laser, matching it in transverse profile and polarization. At the center of the nonlinear crystal, we write the corresponding electric field as

$$\hat{\mathbf{E}}_p = \int_{\mathcal{O}(\omega_p)} d\omega \, \mathrm{i}\sqrt{\frac{\hbar\omega}{\epsilon_0 c\pi}} \boldsymbol{\epsilon}_p A_p(x,y)\hat{p}(\omega)e^{\mathrm{i}kz} + \text{H.c.} \tag{2.9}$$

As usual, $\hat{p}(\omega)$ and $\hat{p}^\dagger(\omega)$ are the annihilation and creation operators for the corresponding modes, and satisfy continuous canonical commutation relations $[\hat{p}(\omega), \hat{p}^\dagger(\omega')] = \delta(\omega - \omega')$.

2.4 Hamiltonian of the system

Now that we have introduced the relevant modes required to describe our system, we are in conditions to provide a physical model that will allow us to describe its evolution. We do this in the form of a Hamiltonian that can be written as the sum of the following four contributions:

1. The Hamiltonian of the cavity modes [18]

$$\hat{H}_{cavity} = \sum_{j=s,i} \hbar\omega_j \hat{a}_j^\dagger \hat{a}_j \tag{2.10}$$

   In the following, we will assume that phase matching has been tuned such that both signal and idler resonate at frequencies close to half the pump frequency, $\omega_p/2 = \omega_0$, as sketched in fig. 2.1.

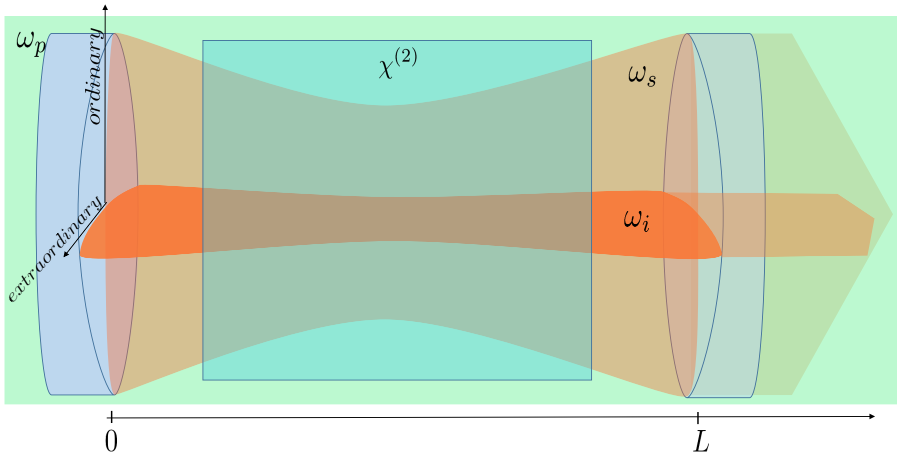

FIGURE 2.2. Sketch of the system under consideration: type II optical parametric oscillator. A cavity contains a crystal with second-order nonlinearity, capable of down-converting photons from a pump at frequency ω_p (blue background), to two cavity modes with frequencies ω_s (light red) and ω_i (dark red) polarized, respectively, along the the ordinary and extraordinary axes of the crystal. In our model, we assume that the cavity mirrors are transparent to the pump frequency.

2. The Hamiltonian for the external modes [18]

$$\hat{H}_{external} = \sum_{j=s,i} \int_{\mathcal{O}(\omega_j)} d\omega\, \hbar\omega\, \hat{b}_j^\dagger(\omega)\hat{b}_j(\omega) + \int_{\mathcal{O}(\omega_p)} d\omega\, \hbar\omega\, \hat{p}^\dagger(\omega)\hat{p}(\omega) \tag{2.11}$$

As mentioned above, these external fields will be excited by a coherent source (laser) at frequencies $2\omega_0$ and ω_0, motivated by the possibility of locking the phase relation between the signal and the idler, as well as their frequencies to the degenerate one (active phase locking [9]).

3. We assume that one of the mirrors is partially transmissive, inducing the following coupling term between the cavity modes and the external field [18]:

$$\hat{H}_{interaction} = \sum_{j=s,i} \int_{\mathcal{O}(\omega_j)} d\omega \sqrt{\frac{\gamma}{\pi}}\, \mathrm{i}\hbar(\hat{b}_j^\dagger(\omega)\hat{a}_j - \mathrm{H.c}). \tag{2.12}$$

Here, γ is the so-called damping rate, which is proportional to the transmissivity of the mirror, assumed independent of the frequency for the considered frequencies.

4. The Hamiltonian of the nonlinear interaction (PDC) [19]:

$$\hat{H}_{PDC} = \int_{\mathcal{O}(2\omega_0)} d\omega \sqrt{\frac{\kappa}{\pi}}\, \mathrm{i}\hbar(\hat{p}^\dagger(\omega)\hat{a}_s\hat{a}_i - \mathrm{H.c}) \tag{2.13}$$

where $\sqrt{\kappa/\pi}$ is proportional to the nonlinear susceptibility, the crystal length, and the overlapping integral between the relevant modes, $\int dxdyA_p^*(x,y)A^2(x,y)$.

2.5 The cavity as an open quantum system

Unlike closed quantum systems, open systems do not follow unitary time evolution because of their interaction with the environment. The system looses and gains energy from the environment. This makes the Schrödinger equation for pure states inadequate to describe the dynamics of an open quantum system, unless the degrees of freedom of the environment are also taken into account. Without the latter, a mixed-state description is required for the system, specifically through a density operator $\hat{\rho}$. For a closed system, the time evolution of the density operator is coherent and given by the Liouville-von Neumann equation, $\mathrm{i}\hbar d\hat{\rho}/dt = [\hat{H},\hat{\rho}]$, which is equivalent to the Schrödinger equation. For open systems, incoherent evolution arises once the degrees of freedom of the environment are effectively integrated (traced out). In this section, we will perform this effective integration of the environmental degrees of freedom of our model (external modes), and find the evolution equation for the density operator of the cavity modes alone. We start by moving to a more convenient picture where only the interaction terms of the Hamiltonian remain.

2.5.1 Interaction picture Hamiltonian

As exposed above, the Hamiltonian of the system under consideration is

$$\begin{aligned}\hat{H} = &\sum_{j=s,i}\hbar\omega_j\hat{a}_j^\dagger\hat{a}_j + \sum_{j=s,i}\int_{\mathcal{O}(\omega_j)} d\omega\,\hbar\omega\,\hat{b}_j^\dagger(\omega)\hat{b}_j(\omega) + \int_{\mathcal{O}(2\omega_0)} d\omega\,\hbar\omega\,\hat{p}^\dagger(\omega)\hat{p}(\omega) \\ &+ \sum_{j=s,i}\int_{\mathcal{O}(\omega_j)} d\omega\sqrt{\frac{\gamma}{\pi}}\,\mathrm{i}\hbar(\hat{b}_j^\dagger(\omega)\hat{a}_j - \mathrm{H.c}) + \int_{\mathcal{O}(2\omega_0)} d\omega\sqrt{\frac{\kappa}{\pi}}\,\mathrm{i}\hbar(\hat{p}^\dagger(\omega)\hat{a}_s\hat{a}_i - \mathrm{H.c}).\end{aligned} \tag{2.14}$$

We first move to a new picture rotating at the frequencies of the injected lasers. This is done by applying the unitary operator $\hat{U}_1 = \exp(\hat{H}_1 t/\mathrm{i}\hbar)$ to the state of the system $\hat{\rho}$, with

$$\hat{H}_1 = \sum_{j=s,i}\hbar\omega_0\hat{a}_j^\dagger\hat{a}_j + \sum_{j=s,i}\int_{\mathcal{O}(\omega_j)} d\omega\,\hbar\omega_0\,\hat{b}_j^\dagger(\omega)\hat{b}_j(\omega) + \int_{\mathcal{O}(2\omega_0)} d\omega\,\hbar 2\omega_0\,\hat{p}^\dagger(\omega)\hat{p}(\omega). \tag{2.15}$$

The transformed state $\tilde{\rho}_1 = \hat{U}_1^\dagger\hat{\rho}\hat{U}_1$, evolves according to the transformed Hamiltonian $\tilde{H}_1 = \hat{U}_1^\dagger\hat{H}\hat{U}_1 - \hat{H}_1$, which noting that $\hat{U}_1^\dagger\hat{H}\hat{U}_1 = \hat{H}$, means that the Hamiltonian in the rotating picture can be written as

$$\begin{aligned}\tilde{H}_1 = &\sum_{j=s,i}\hbar\Delta_j\hat{a}_j^\dagger\hat{a}_j + \sum_{j=s,i}\int_{\mathcal{O}(\omega_j)} d\omega\hbar\Delta_b(\omega)\,\hat{b}_j^\dagger(\omega)\hat{b}_j(\omega) + \int_{\mathcal{O}(2\omega_0)} d\omega\,\hbar\Delta_p(\omega)\,\hat{p}^\dagger(\omega)\hat{p}(\omega) \\ &+ \sum_{j=s,i}\int_{\mathcal{O}(\omega_j)} d\omega\sqrt{\frac{\gamma}{\pi}}\,\mathrm{i}\hbar(\hat{b}_j^\dagger(\omega)\hat{a}_j - \mathrm{H.c.}) + \int_{\mathcal{O}(2\omega_0)} d\omega\mathrm{i}\hbar\sqrt{\frac{\kappa}{\pi}}\,(\hat{p}^\dagger(\omega)\hat{a}_s\hat{a}_i - \mathrm{H.c}),\end{aligned} \tag{2.16}$$

where we have introduced the detunings

$$\Delta_j = \omega_j - \omega_0, \quad \Delta_b(\omega) = \omega - \omega_0, \quad \Delta_p(\omega) = \omega - 2\omega_0. \tag{2.17}$$

In the absence of lasers, the environmental modes are in a thermal state. In contrast, in our case $\langle \hat{b}_j(\omega)\rangle \neq 0$ and $\langle \hat{p}(\omega)\rangle \neq 0$ for the frequencies pumped by the lasers. Hence, it is convenient to move to a picture where the environments are in a thermal state, what can be done by applying the displacement unitary operator $\hat{D} = \exp\big(\sum_{j=s,i}[\alpha_j \hat{b}_j^\dagger(\omega_0) - \alpha_j^* \hat{b}_j(\omega_0)] + [\alpha_p \hat{p}^\dagger(2\omega_0) - \alpha_p^* \hat{p}(2\omega_0)]\big)$ to the state. Here, the amplitudes α_j characterize the strength of the pumping (proportional to the square root of the corresponding laser power), with a phase given by that of the corresponding laser. This operator, as its name suggests, displaces any Gaussian state with all its statistical attributes by a magnitude of α_j in phase space. The transformed state $\tilde{\rho}_D = \hat{D}^\dagger \tilde{\rho}_1 \hat{D}$ evolves then according to the Hamiltonian

$$\tilde{H}_D = \hat{D}^\dagger \tilde{H}_1 \hat{D} = \tilde{H}_1 + \mathrm{i}\hbar \sum_{j=i,s} (\epsilon_j \hat{a}_j^\dagger - \epsilon_j^* \hat{a}_j) + \mathrm{i}\hbar(\epsilon_p \hat{a}_i^\dagger \hat{a}_s^\dagger - \epsilon_p^* \hat{a}_i \hat{a}_s), \tag{2.18}$$

where

$$\epsilon_j = -\sqrt{\gamma/\pi}\,\alpha_j, \epsilon_p = \sqrt{\kappa/\pi}\,\alpha_p. \tag{2.19}$$

A final change of picture effected by the unitary operator $\hat{U}_2 = \exp(\hat{H}_2 t/\mathrm{i}\hbar)$, is done to keep only the interaction term of the Hamiltonian, therefore choosing

$$\begin{aligned}\hat{H}_2 = \sum_{j=s,i} \hbar\Delta_j \hat{a}_j^\dagger \hat{a}_j + \sum_{j=s,i} \int_{\mathcal{O}(\omega_j)} d\omega \hbar\Delta_b\, \hat{b}_j^\dagger(\omega)\hat{b}_j(\omega) + \int_{\mathcal{O}(2\omega_0)} d\omega\, \hbar\Delta_p\, \hat{p}^\dagger(\omega)\hat{p}(\omega) \\ + \mathrm{i}\hbar \sum_{j=i,s} (\epsilon_j a_j^\dagger - \epsilon_j^* a_j) + \mathrm{i}\hbar(\epsilon_p a_i^\dagger a_s^\dagger - \epsilon_p^* a_i a_s).\end{aligned} \tag{2.20}$$

We define transformed intra-cavity operators $\tilde{a}_j(t) = \hat{U}_2^\dagger(t)\hat{U}_1^\dagger(t)\hat{a}_j \hat{U}_1(t)\hat{U}_2(t)$. On the other hand, the external mode's bosonic operators change as $\hat{U}_2^\dagger \hat{U}_1^\dagger \hat{b}(\omega)\hat{U}_1\hat{U}_2 = \hat{b}(\omega)e^{-\mathrm{i}\omega t}$ and $\hat{U}_2^\dagger \hat{U}_1^\dagger \hat{p}(\omega)\hat{U}_1\hat{U}_2 = \hat{p}(\omega)e^{-\mathrm{i}\omega t}$. With this in mind, the transformed state $\tilde{\rho} = \hat{U}_2^\dagger \tilde{\rho}_D \hat{U}_2$ evolves according to the Hamiltonian

$$\begin{aligned}\tilde{H}_{int} &= \hat{U}_2^\dagger \tilde{H}_D \hat{U}_2 - \hat{H}_2 \\ &= \sum_{j=s,i} \int_{-\infty}^{\infty} d\omega \mathrm{i}\hbar \sqrt{\frac{\gamma}{\pi}}\, \hat{b}_j^\dagger(\omega) e^{\mathrm{i}\omega t} \tilde{a}_j(t) + \int_{-\infty}^{\infty} d\omega \mathrm{i}\hbar \sqrt{\frac{\kappa}{\pi}}\, \hat{p}^\dagger(\omega) e^{\mathrm{i}\omega t} \tilde{a}_s(t)\tilde{a}_i(t) - \mathrm{H.C.}\end{aligned} \tag{2.21}$$

2.5.2 Master Equation

As mentioned above, the time evolution of density operator of an open system is not unitary. We are now going to show this explicitly in our system by deriving a new time evolution of the reduced density operator of the cavity modes by tracing out the external modes. We start from the Liouville-von Neumann equation for the state of the system (cavity modes) and the environment (external modes) in the transformed picture, which reads

$$\frac{d\tilde{\rho}}{dt} = \frac{1}{\mathrm{i}\hbar}[\tilde{H}_{int}, \tilde{\rho}] \tag{2.22}$$

Since we are only interested in the dynamics of the system, we will trace over the environment. Without further approximations, this wouldn't bring us too far. Hence, in the following we will seek some reasonable approximations that will allow us to achieve our goal.

Let us first consider a very raw approximation, consisting on the assumption that the composite system is separable, that is, $\tilde{\rho}(t) = \tilde{\rho}_c(t) \otimes \hat{\rho}_e(t)$ where $\tilde{\rho}_c(t)$ and $\hat{\rho}_e(t)$ are, respectively, the density operators of the system and the environment. Note that by doing this we are completely ignoring any correlations between the system and the environment at all times. Introducing this in the Liouville-von Neumann equation, and tracing over the environment, we get

$$\frac{d}{dt}\mathrm{tr}_{\mathrm{ext}}\{\tilde{\rho}_c(t) \otimes \hat{\rho}_e(t)\} = \frac{d\tilde{\rho}_c(t)}{dt} = \frac{1}{\mathrm{i}\hbar}\Big(\underbrace{\mathrm{tr}_{\mathrm{ext}}\{\tilde{H}_{int}\hat{\rho}_e(t)\}}_{\hat{H}_{eff}}\tilde{\rho}_c(t) - \tilde{\rho}_c(t)\underbrace{\mathrm{tr}_{\mathrm{ext}}\{\tilde{H}_{int}\hat{\rho}_e(t)\}}_{\hat{H}_{eff}}\Big). \tag{2.23}$$

As can be seen from the above equation, the evolution of the system is again unitary. So by considering the composite system to be separable we are completely ignoring the incoherent processes expected between the system and the environment. Let us now see what the effective Hamiltonian looks like, for what we need the statistical attributes of the external modes, so that we can take the trace. Consider an environment in thermal equilibrium at some temperature T, characterized by a density operator [11]

$$\hat{\rho}_e = \frac{e^{-\hat{H}_{external}/k_B T}}{\mathrm{tr}\{e^{-\hat{H}_{external}/k_B T}\}} \equiv \bar{\rho}_{\mathrm{th}}, \tag{2.24}$$

further assumed to remain unaffected by the interaction with the system. Since this thermal state has a Gaussian distribution, it can be characterized by first and second order moments, which are

$$\begin{aligned} \mathrm{tr}_{\mathrm{ext}}\{\hat{b}(\omega)\bar{\rho}_{\mathrm{th}}\} &= 0 & \mathrm{tr}_{\mathrm{ext}}\{\hat{b}^\dagger(\omega)\bar{\rho}_{\mathrm{th}}\} &= 0 \\ \mathrm{tr}_{\mathrm{ext}}\{\hat{b}(\omega)\hat{b}(\omega')\bar{\rho}_{\mathrm{th}}\} &= 0 & \mathrm{tr}_{\mathrm{ext}}\{\hat{b}^\dagger(\omega)\hat{b}^\dagger(\omega')\bar{\rho}_{\mathrm{th}}\} &= 0 \\ \mathrm{tr}_{\mathrm{ext}}\{\hat{b}^\dagger(\omega)\hat{b}(\omega')\bar{\rho}_{\mathrm{th}}\} &= \bar{n}\delta(\omega-\omega') & \mathrm{tr}_{\mathrm{ext}}\{\hat{b}(\omega)\hat{b}^\dagger(\omega')\bar{\rho}_{\mathrm{th}}\} &= (\bar{n}+1)\delta(\omega-\omega'), \end{aligned} \tag{2.25}$$

where we have made the approximation that all the frequencies have the same thermal occupation $\bar{n}$, very good approximation in optical systems. Now we can take the trace over the environment to find the effective Hamiltonian as

$$\hat{H}_{eff} = \mathrm{tr}_{\mathrm{ext}}\{\tilde{H}_{int}\hat{\rho}_e\} = 0. \tag{2.26}$$

As $\tilde{H}_{int}$ is linear in bosonic environmental operators, the above expression is a function of only first order moments of the thermal state, which are zero. So the rate of change in the system due to interaction with the environment is zero for the above approximations. The obvious reason for this is that our approximations are not good enough for the environment under consideration, and we need to consider a higher degree of correlation between the system and the environment.

In order to do this, we first recast the von Neumann equation (2.22) into a new form more suited for further approximations. This equation has the formal solution

$$\tilde{\rho}(t) = \tilde{\rho}(0) + \frac{1}{\mathrm{i}\hbar}\int_0^t dt'[\tilde{H}_{int}(t'), \tilde{\rho}(t')], \tag{2.27}$$

Reintroducing this expression into the von Neumann equation, we obtain

$$\frac{d\tilde{\rho}(t)}{dt} = \frac{1}{\mathrm{i}\hbar}[\tilde{H}_{int}(t), \tilde{\rho}(0)] - \frac{1}{\hbar^2}\int_0^t dt'[\tilde{H}_{int}(t), [\tilde{H}_{int}(t'), \tilde{\rho}(t')]]. \tag{2.28}$$

We make the partial trace over the external modes, obtaining

$$\frac{d\tilde{\rho}_c(t)}{dt} = \frac{1}{\mathrm{i}\hbar}\mathrm{tr}_{\mathrm{ext}}\{[\tilde{H}_{int}(t), \tilde{\rho}(0)]\} - \frac{1}{\hbar^2}\int_0^t dt'\mathrm{tr}_{\mathrm{ext}}\{[\tilde{H}_{int}(t), [\tilde{H}_{int}(t'), \tilde{\rho}(t')]]\}. \tag{2.29}$$

Next we apply the following approximations to this new form of von Neumann equation:

- The density operator of the environment does not change in time i.e. $\rho_e(0) = \rho_e(t)\forall t$ and it is in the thermal state $\bar{\rho}_{th}$ provided above.
- The composite system is separable, which means that the system and environment are not correlated. Hence, the density operator can be factorized as $\tilde{\rho}(t) = \tilde{\rho}_c(t) \otimes \bar{\rho}_{\mathrm{th}}$. This is known as the **Born approximation**.
- We assume that the evolution of the system at a given time depends only on the state of the system at that time, which is known as **Markov approximation**. In fact, when the interaction rates between system and environment do not depend on the frequency of the environmental modes, as is our case, it is easy to see that the previous approximations directly imply this Markov approximation.

Applying the Born approximation, taking the trace over environment and making the variable change $t' = t - \tau$ in the time integral, we get the following integro-differential equation for the reduced density operator:

$$\frac{d\tilde{\rho}_c(t)}{dt} = -\frac{1}{\hbar^2}\int_0^t d\tau\mathrm{tr}_{\mathrm{ext}}\{[\tilde{H}_{int}(t), [\tilde{H}_{int}(t-\tau), \tilde{\rho}_c(t-\tau) \otimes \bar{\rho}_{\mathrm{th}}]]\}. \tag{2.30}$$

Now by applying the Markov approximation $\tilde{\rho}(t-\tau) = \tilde{\rho}(t)$ we get master equation in Born-Markov approximation:

$$\frac{d\tilde{\rho}_c(t)}{dt} = -\frac{1}{\hbar^2}\int_0^t d\tau\mathrm{tr}_{\mathrm{ext}}\{[\tilde{H}_{int}(t), [\tilde{H}_{int}(t-\tau), \tilde{\rho}_c(t) \otimes \bar{\rho}_{\mathrm{th}}]]\}. \tag{2.31}$$

Even though the interaction Hamiltionan looks complicated, it can be considered as a linear combination of three terms of the same type

$$\tilde{H}_{int} = \mathrm{i}\hbar \sum_{j=i,s,p} C_j \int_{-\infty}^{+\infty} d\omega\, \hat{b}_j^\dagger(\omega) e^{\mathrm{i}\omega t} \hat{J}_j(t) - H.C, \tag{2.32}$$

where $C_{i,s} = \sqrt{\gamma/\pi}, C_p = \sqrt{\kappa/\pi}, \hat{b}_p^\dagger(\omega) = \hat{p}^\dagger(\omega)$, $\hat{J}_i = \tilde{a}_i$, $\hat{J}_s = \tilde{a}_s$, and $\hat{J}_p = \tilde{a}_s\tilde{a}_i$. We can treat each term in the summation of (2.32) separately and add them up at the end. So for now we ditch the summation and define a new Hamiltonian:

$$\tilde{H} = \mathrm{i}\hbar C \int_{-\infty}^{+\infty} d\omega\, \hat{b}^\dagger(\omega) e^{\mathrm{i}\omega t} \hat{J}(t) - \mathrm{H.C.} \tag{2.33}$$

(2.30) becomes

$$\begin{aligned}\frac{d\tilde{\rho}_c(t)}{dt} = -\frac{\mathrm{i}C^2}{\hbar} \int_0^t d\tau (&\mathrm{tr}_{\mathrm{ext}}\{\tilde{H}(t)\tilde{H}(t-\tau)\tilde{\rho}(t)\} - \mathrm{tr}_{\mathrm{ext}}\{\tilde{H}(t)\tilde{\rho}(t)\tilde{H}(t-\tau)\} \\ &- \mathrm{tr}_{\mathrm{ext}}\{\tilde{H}(t-\tau)\tilde{\rho}(t)\tilde{H}(t)\} + \mathrm{tr}_{\mathrm{ext}}\{\tilde{\rho}(t)\tilde{H}(t-\tau)\tilde{H}(t)\}).\end{aligned} \tag{2.34}$$

Because of (2.25) all the terms with $\hat{b}(\omega)\hat{b}(\omega')$ and $\hat{b}^\dagger(\omega)\hat{b}^\dagger(\omega')$ can be set to zero, for example

$$C^2 \int_0^t d\tau \int_{-\infty}^{+\infty} d\omega \int_{-\infty}^{+\infty} d\omega' \hat{J}(t)\hat{J}(t-\tau)\tilde{\rho}_c(t) e^{\mathrm{i}\omega t} e^{\mathrm{i}\omega'(t-\tau)} \mathrm{tr}_{\mathrm{ext}}\{\hat{b}^\dagger(\omega)\hat{b}^\dagger(\omega'\bar{\rho}_{\mathrm{th}}\} = 0, \tag{2.35}$$

so that only terms which have $\hat{b}(\omega)\hat{b}^\dagger(\omega')$ or $\hat{b}^\dagger(\omega)\hat{b}(\omega')$ remain, for example

$$\begin{aligned}C^2 \int_0^t d\tau \int_{-\infty}^{+\infty} d\omega \int_{-\infty}^{+\infty} & d\omega' \hat{J}(t)\hat{J}^\dagger(t-\tau)\tilde{\rho}_c(t) e^{\mathrm{i}\omega t} e^{-\mathrm{i}\omega'(t-\tau)} \mathrm{tr}_{\mathrm{ext}}\{\hat{b}(\omega)\hat{b}^\dagger(\omega')\bar{\rho}_{\mathrm{th}}\} \\ &= C^2 \int_0^t d\tau \hat{J}(t)\hat{J}^\dagger(t-\tau)\tilde{\rho}_c(t) \underbrace{\int_{-\infty}^{+\infty} d\omega e^{\mathrm{i}\omega\tau}}_{2\pi\delta(\tau)} = C^2\pi\bar{n}\hat{J}(t)\hat{J}^\dagger(t)\tilde{\rho}_c(t),\end{aligned} \tag{2.36}$$

Here we have made use of the property $\int_{\tau_0}^{\tau_1} d\tau f(\tau)\delta(\tau-\tau_0) = f(\tau_0)/2$ of the Dirac delta. Note that the delta function appears only because we could extend the limits of the integral from $-\infty$ to ∞ based on rotating wave approximation and taking a frequency independent coupling constant. This same Dirac delta is responsible for the Markov condition to appear naturally without having to force it.

After taking care of all 16 terms in (2.34), we end up with the equation

$$\frac{d\tilde{\rho}_c}{dt} = (\bar{n}+1)C^2\pi\left(2\hat{J}\tilde{\rho}_c\hat{J}^\dagger - \hat{J}^\dagger\hat{J}\tilde{\rho}_c - \tilde{\rho}_c\hat{J}^\dagger\hat{J}\right) + \bar{n}C^2\pi\left(2\hat{J}^\dagger\tilde{\rho}_c\hat{J} - \hat{J}\hat{J}^\dagger\tilde{\rho}_c - \tilde{\rho}_c\hat{J}\hat{J}^\dagger\right), \tag{2.37}$$

This equation is in the so-called Lindblad form and can be written as

$$\frac{d\tilde{\rho}_c}{dt} = \sum_n \kappa_n \left(2\hat{J}_n\tilde{\rho}\hat{J}_n^\dagger - \hat{J}_n^\dagger\hat{J}_n\tilde{\rho} - \tilde{\rho}\hat{J}_n^\dagger\hat{J}_n\right) = \sum_n \kappa_n \mathcal{D}_{J_n}[\tilde{\rho}], \tag{2.38}$$

with $\{\kappa_1 = (\bar{n}+1)C^2\pi, \hat{J}_1 = \hat{J}\}$ and $\{\kappa_2 = \bar{n}C^2\pi, \hat{J}_2 = \hat{J}^\dagger\}$. $\hat{J}_n$ are called a jump operators, because they are responsible for the incoherent jumps in the evolution of the density operator. Now bringing back the summation from (2.32) we get

$$\frac{d\tilde{\rho}_c}{dt} = \gamma\mathcal{D}_{\tilde{a}_i}[\tilde{\rho}_c] + \gamma\mathcal{D}_{\tilde{a}_s}[\tilde{\rho}_c] + \kappa\mathcal{D}_{\tilde{a}_s\tilde{a}_i}[\tilde{\rho}_c], \tag{2.39}$$

where we have set the temperature of the environment to zero ($\bar{n} = 0$), a very good approximation for optical frequencies and room temperature.

Finally we come back to the original picture, except for the initial transformation $\hat{U}_1(t)$, which means that we stay in a picture rotating at frequency ω_0, as otherwise the Hamiltonian becomes time dependent, which is inconvenient. The master equation governing the evolution of the density operator of the cavity modes in this picture, which we simply denote by $\hat{\rho}$ from now on, reads

$$\frac{d\hat{\rho}}{dt} = -\mathrm{i}\Big[\hat{H}_0 + \hat{H}_{inj} + \hat{H}_{PDC}, \hat{\rho}\Big] + \gamma\mathcal{D}_{\hat{a}_i}[\hat{\rho}] + \gamma\mathcal{D}_{\hat{a}_s}[\hat{\rho}] + \kappa\mathcal{D}_{\hat{a}_s\hat{a}_i}[\hat{\rho}], \tag{2.40}$$

with

$$\hat{H}_0 = \sum_{j=s,i} \Delta_j \hat{a}_j^\dagger \hat{a}_j, \tag{2.41a}$$

$$\hat{H}_{inj} = \sum_{j=s,i} \mathrm{i}(\epsilon_j a_j^\dagger - \epsilon_j^* a_j), \tag{2.41b}$$

$$\hat{H}_{PDC} = \mathrm{i}(\epsilon_p a_i^\dagger a_s^\dagger - \epsilon_p^* a_i a_s). \tag{2.41c}$$

2.5.3 Classical limit of the system

From the master equation we can find the dynamics of the system within the classical limit. This is done by assuming the system to be in a coherent state $\hat{\rho}_{coh}(t) = |\alpha_s(t), \alpha_i(t)\rangle\langle\alpha_s(t), \alpha_i(t)|$ at all times and finding the evolution of the expectation value of the annihilation operators $\mathrm{tr}\{\hat{a}_j\hat{\rho}_{coh}(t)\} = \alpha_j(t)$, which correspond to the complex classical phase-space variables of our cavity modes. By considering the system to be in coherent state we are ignoring the noisy nature of our quantum system (except for vacuum fluctuations) as will be apparent in Chapter 4. We choose the coherent state because, by construction, it is the quantum state providing statistics that are closest to what can be expected from a classical system [11, 18].

Consider the evolution equation of the expectation value of any operator $\hat{A}$:

$$\begin{aligned}\frac{d\mathrm{tr}\{\hat{A}\hat{\rho}\}}{dt} &= \mathrm{tr}\left\{\hat{A}\frac{d\hat{\rho}}{dt}\right\} \\ &= \mathrm{tr}\left\{-\mathrm{i}\hat{A}[\hat{H}_0 + \hat{H}_{inj} + \hat{H}_{PDC}, \hat{\rho}] + \gamma\hat{A}\mathcal{D}_{\hat{a}_i}[\hat{\rho}] + \gamma\hat{A}\mathcal{D}_{\hat{a}_s}[\hat{\rho}] + \kappa\hat{A}\mathcal{D}_{\hat{a}_s\hat{a}_i}[\hat{\rho}]\right\}\end{aligned} \tag{2.42}$$

Using the cyclic property of the trace we can rewrite the expression in terms of commutators. For example, for the Hamiltonian terms we obtain

$$\mathrm{tr}\{\hat{A}[\hat{H},\rho]\} = \mathrm{tr}\{\hat{A}\hat{H}\rho\} - \mathrm{tr}\{\hat{A}\rho\hat{H}\} = \mathrm{tr}\{\hat{A}\hat{H}\rho\} - \mathrm{tr}\{\hat{H}\hat{A}\rho\} = \langle[\hat{A},\hat{H}]\rangle \tag{2.43}$$

Similarly we can do the same to the terms in Lindblad form

$$\begin{aligned}
\mathrm{tr}\{\hat{A}\mathcal{D}_{\hat{J}}[\hat{\rho}]\} =& 2\mathrm{tr}\{\hat{A}\hat{J}\hat{\rho}\hat{J}^\dagger\} - \mathrm{tr}\{\hat{A}\hat{J}^\dagger\hat{J}\hat{\rho}\} - tr\{\hat{A}\hat{\rho}\hat{J}^\dagger\hat{J}\} \\
=& \mathrm{tr}\{\hat{A}\hat{J}\hat{\rho}\hat{J}^\dagger - \hat{A}\hat{J}^\dagger\hat{J}\hat{\rho}\} + \mathrm{tr}\{\hat{A}\hat{J}\hat{\rho}\hat{J}^\dagger - \hat{A}\hat{\rho}\hat{J}^\dagger\hat{J}\} \\
=& \mathrm{tr}\{(\hat{J}^\dagger\hat{A} - \hat{A}\hat{J}^\dagger)\hat{J}\hat{\rho}\} + \mathrm{tr}\{\hat{J}^\dagger(\hat{A}\hat{J} - \hat{J}\hat{A})\hat{\rho}\} \\
=& \langle[\hat{J}^\dagger,\hat{A}]\hat{J}\rangle + \langle\hat{J}^\dagger[\hat{A},\hat{J}]\rangle
\end{aligned} \tag{2.44}$$

Using these expressions, and the fact that the coherent states are right (left) eigenstates of the annihilation (creation) operators, we easily find the classical equations of the system:

$$\dot{\alpha}_s = \epsilon_s - (\gamma + \mathrm{i}\Delta_s + \kappa|\alpha_i|^2)\alpha_s + \epsilon_p\alpha_i^*, \tag{2.45a}$$

$$\dot{\alpha}_i = \epsilon_i - (\gamma + \mathrm{i}\Delta_i + \kappa|\alpha_s|^2)\alpha_i + \epsilon_p\alpha_s^*. \tag{2.45b}$$

CHAPTER 3

Phase-space representation

In order to make sense of the statistical nature of the density operator used in the previous chapter, it is convenient to have a tool that gives some kind of distribution in phase space that we can visualize. We strive for a phase space representation because the uncertainty principle provides bounds on the minimum area a phase space distribution can have (quantum noise). In this chapter we introduce such phase space representations, and explain how to find their evolution equation for a given master equation, developing the method explicitly for our system.

3.1 Phase-space quasiprobability distributions

Let us start by drawing some inspiration from classical probability theory. The Fourier transform of a probability distribution $P(x, p)$ in the phase space of a single mode can be seen as the average value of the exponential Fourier transform Kernel:

$$\int_{\mathbb{R}^2} dxdp \; e^{\mathrm{i}(vp-ux)} P(x,p) = \langle e^{\mathrm{i}(vp-ux)} \rangle, \tag{3.1}$$

where in the following we use indistinctly $\langle \cdot \rangle$ to denote classical or quantum expectation values, since from the context it will be evident which one we are referring to. This object is known as characteristic function, and it contains the same amount of information as probability distribution (as it is the same function represented in the Fourier domain).

We can build a similar quantum mechanical expression, simply by replacing the classical phase-space variables (x, p) by the corresponding quantum operators $(\hat{x}, \hat{p})$, and interpreting the expectation value as a quantum expectation value, that is,

$$\langle e^{\mathrm{i}(v\hat{p}-u\hat{x})} \rangle = \mathrm{tr}\{\hat{\rho} e^{\mathrm{i}(v\hat{p}-u\hat{x})}\}. \tag{3.2}$$

Note that we use the definitions $\hat{x} = \hat{a}^\dagger + \hat{a}$ and $\hat{p} = \mathrm{i}(\hat{a}^\dagger - \hat{a})$ for the position and momentum quadrature operators, so that they satisfy the commutation relation $[\hat{x}, \hat{p}] = 2\mathrm{i}$.

Instead of working with position and momentum, it is convenient to use a so-called complex representation based on annihilation and creation operators. Using the relation between these operators, the quantum characteristic function introduced above takes the form $\langle \hat{D}(\beta) \rangle$, where $\hat{D}(\beta) = \exp(\beta \hat{a}^\dagger - \beta^* \hat{a})$ is the displacement operator, and now the characteristic function depends on a complex variable β, instead of the real ones (u, v). Now, since $\hat{a}^\dagger$ and $\hat{a}$ do not commute, alternative forms of the quantum characteristic function such as $\langle \exp(\beta \hat{a}^\dagger) \exp(-\beta^* \hat{a}) \rangle$ or $\exp(-\beta^* \hat{a}) \langle \exp(\beta \hat{a}^\dagger) \rangle$ are inequivalent. On the other hand, all of them are equally reasonable, since their classical counterparts converge. Therefore, this inspires a more general definition of the quantum mechanical characteristic function through a generalized displacement operator

$$\hat{D}_z(\beta) = e^{\beta \hat{a}^\dagger - \beta^* \hat{a}} e^{z|\beta|^2} \tag{3.3}$$

where z can take any value between -1 and 1. This generalized characteristic function takes the form $\chi_z(\beta) = \langle \hat{D}_z(\beta) \rangle$, such that the quantum counterpart of (3.1) becomes

$$\chi_z(\beta) = \mathrm{tr}\{\hat{\rho} e^{\beta \hat{a}^\dagger - \beta^* \hat{a}} e^{z|\beta|^2}\} = \int_{\mathbb{C}} d^2\alpha e^{\beta \alpha^* - \beta^* \alpha} F_z(\alpha). \tag{3.4}$$

$F_z(\alpha)$ is the quantum mechanical phase-space distribution we were seeking for. Note that we get an expression for an infinite number of probability distributions, for different values of z, all with the same amount of information about the statistical attributes of the system. The form of these distributions is found by inverting the Fourier transform relation above, leading to

$$F_z(\alpha) = \int_{\mathbb{C}} \frac{d^2\beta}{\pi^2} e^{\beta^* \alpha - \beta \alpha^*} \chi_z(\beta), \tag{3.5}$$

where π^2 is the normalizing factor. Depending on the problem at hand we can choose any of the distributions. Even though they are real and normalized to one, as we will see next, except for the choice $z = -1$, none of the distributions satisfy all the axioms of a classical probability distribution (e.g., some can be negative or have strong divergences). These are then not probability density functions in the classical sense, and they are usually called quasiprobability distributions in quantum optics. Nevertheless we can use these functions to get some insight into the state and dynamics of quantum systems. In the rest of the chapter we will briefly introduce these quasiprobability distributions, and explain how to find their dynamics.

The choices $z = 0, 1,$ and -1 are specially useful and well studied. The corresponding distributions where found independently before a general formulation was established, and receive special names: $F_0(\alpha) \equiv W(\alpha)$ is called Wigner function, $F_1(\alpha) \equiv P(\alpha)$

is called Glauber-Sudarshan distribution, and $F_{-1}(\alpha) \equiv Q(\alpha)$ is called Husimi distribution. All three quasiprobability distributions can be used to calculate quantum expectation values, but connect with different orderings of annihilation and creation operators. Specifically, the Wigner function allows for symmetrically ordered expectation values $\langle (\hat{a}^{\dagger m}\hat{a}^n)^s\rangle = \int_{\mathbb{C}} d^2\alpha W(\alpha)\alpha^{*m}\alpha^n$, the Glauber-Sudarshan for normally ordered ones $\langle \hat{a}^{\dagger m}\hat{a}^n\rangle = \int_{\mathbb{C}} d^2\alpha P(\alpha)\alpha^{*m}\alpha^n$, and the Husimi for antinormally ordered ones $\langle \hat{a}^n\hat{a}^{\dagger m}\rangle = \int_{\mathbb{C}} d^2\alpha Q(\alpha)\alpha^{*m}\alpha^n$.

It is possible to show that the Wigner and Glauber-Sudarshan distributions can be negative, with the later showing strong divergences even for simple Gaussian states. On the other hand, the Husimi distribution is always positive, but its connection with antinormally ordered moments makes it not directly connected to a useful physical picture. Hence, the three of them have strengths and weaknesses.

If more than one mode exists, as is our case, the definitions above are trivially generalized. For example, dealing with signal and idler, and collecting their complex amplitudes in $\boldsymbol{\beta} = (\beta_s, \beta_i)^{\mathrm{T}}$, we have

$$\hat{D}_z(\boldsymbol{\beta}) = e^{\beta_s\hat{a}_s^\dagger - \beta_s^*\hat{a}_s + \beta_i\hat{a}_i^\dagger - \beta_i^*\hat{a}_i} e^{z|\beta_s|^2 + z|\beta_i|^2}, \tag{3.6}$$

leading to a characteristic function $\chi_z(\boldsymbol{\beta}) = \langle \hat{D}_z(\boldsymbol{\beta})\rangle$ and a phase-space quasiprobability distribution

$$F_z(\boldsymbol{\alpha}) = \int_{\mathbb{C}^2} \frac{d^4\boldsymbol{\beta}}{\pi^4} e^{\boldsymbol{\beta}^\dagger\boldsymbol{\alpha} - \boldsymbol{\alpha}^\dagger\boldsymbol{\beta}} \chi_z(\boldsymbol{\beta}), \tag{3.7}$$

where again we collected the complex amplitudes as $\boldsymbol{\alpha} = (\alpha_s, \alpha_i)^{\mathrm{T}}$. Note that, from now on, the superscript 'T' denotes the 'transpose'.

3.2 Dynamics of the quasiprobability distribution

We know the evolution equation for the state $\hat{\rho}$ of the cavity, the master equation (2.40). From this, we can derive an equation for the time evolution of the quasiprobability distribution $F_z(\boldsymbol{\alpha})$. This is done in order to find some kind of classical correspondence for the dynamics of the system.

The starting point for the derivation is the time derivative of (3.7), which gives

$$\partial_t F_z(\boldsymbol{\alpha}) = \int_{\mathbb{C}^2} \frac{d^4\boldsymbol{\beta}}{\pi^4} e^{\boldsymbol{\beta}^\dagger\boldsymbol{\alpha} - \boldsymbol{\alpha}^\dagger\boldsymbol{\beta}} \mathrm{tr}\left\{\frac{d\hat{\rho}}{dt}\hat{D}_z(\boldsymbol{\beta})\right\}. \tag{3.8}$$

We can substitute $d\hat{\rho}/dt$ by the master equation (2.40), and then use the cyclic property of the trace to make the Hamiltonian and jump operators act on $\hat{D}_z(\boldsymbol{\beta})$. Then, the trick consists on using the following identities:

$$\hat{a}_j^\dagger \hat{D}_z(\beta) = \left(\partial_{\beta_j} - \frac{(z-1)}{2}\beta_j^*\right)\hat{D}_z(\beta), \tag{3.9a}$$

$$\hat{a}_j \hat{D}_z(\beta) = \left(-\partial_{\beta_j^*} + \frac{(z+1)}{2}\beta_j\right)\hat{D}_z(\beta), \tag{3.9b}$$

$$\hat{D}_z(\beta)\hat{a}_j = \left(-\partial_{\beta_j^*} + \frac{(z-1)}{2}\beta_j\right)\hat{D}_z(\beta), \tag{3.9c}$$

$$\hat{D}_z(\beta)\hat{a}_j^\dagger = \left(\partial_{\beta_j} - \frac{(z+1)}{2}\beta_j^*\right)\hat{D}_z(\beta), \tag{3.9d}$$

which are trivial to prove just by taking the corresponding derivative of the displacement operator written in normal or antinormal form:

$$\hat{D}_z(\boldsymbol{\beta}) = e^{\beta_s \hat{a}_s^\dagger + \beta_i \hat{a}_i^\dagger} e^{-\beta_s^* \hat{a}_s - \beta_i^* \hat{a}_i} e^{(z-1)(|\beta_s|^2+|\beta_i|^2)} = e^{-\beta_s^* \hat{a}_s - \beta_i^* \hat{a}_i} e^{\beta_s \hat{a}_s^\dagger + \beta_i \hat{a}_i^\dagger} e^{(z+1)(|\beta_s|^2+|\beta_i|^2)}. \tag{3.10}$$

Note that we are using the notation $\partial_x = \partial/\partial x$ for convenience. Using the equations above, we turn operators into phase-space derivatives, which ultimately turn the master equation into a partial differential equation for the distribution $F_z(\boldsymbol{\alpha})$. For example, we find the following correspondence between terms in the master equation and terms in the partial differential equation of the distribution:

$$\begin{aligned}
\hat{a}_j\hat{\rho} \to \int_{\mathbb{C}^2} \frac{d^4\boldsymbol{\beta}}{\pi^4} e^{\boldsymbol{\beta}^\dagger\boldsymbol{\alpha}-\boldsymbol{\alpha}^\dagger\boldsymbol{\beta}} \mathrm{tr}\{\hat{\rho}\hat{D}_z(\boldsymbol{\beta})\hat{a}_j\} &= \int_{\mathbb{C}^2} \frac{d^4\boldsymbol{\beta}}{\pi^4} e^{\boldsymbol{\beta}^\dagger\boldsymbol{\alpha}-\boldsymbol{\alpha}^\dagger\boldsymbol{\beta}} \big(-\partial_{\beta_j^*} + \frac{(z-1)}{2}\beta_j\big)\chi_z(\boldsymbol{\beta}) \\
&= \int_{\mathbb{C}^2} \frac{d^4\boldsymbol{\beta}}{\pi^4} \pi^2 \chi_z(\boldsymbol{\beta}) \big(\partial_{\beta_j^*} + \frac{(z-1)}{2}\beta_j\big) e^{\boldsymbol{\beta}^\dagger\boldsymbol{\alpha}-\boldsymbol{\alpha}^\dagger\boldsymbol{\beta}} \\
&= \big(\alpha_j - \frac{(z-1)}{2}\partial_{\alpha_j^*}\big)F_z(\boldsymbol{\alpha}),
\end{aligned} \tag{3.11}$$

where we have used the cyclic property of trace and the identity $\int_{\mathbb{C}^2} \frac{d^4\boldsymbol{\beta}}{\pi^4} f(\boldsymbol{\beta})\partial_{\beta_j^*} g(\boldsymbol{\beta}) = -\int_{\mathbb{C}^2} \frac{d^4\boldsymbol{\beta}}{\pi^4} g(\boldsymbol{\beta})\partial_{\beta_j^*} f(\boldsymbol{\beta})$, assuming the product of the functions to vanish at infinity. Similarly, one finds the following correspondences:

$$\rho\hat{a}_j \to \big(\alpha_j - \frac{(z+1)}{2}\partial_{\alpha_j^*}\big)F_z(\boldsymbol{\alpha}), \tag{3.12a}$$

$$\hat{a}_j^\dagger\rho \to \big(\alpha_j^* - \frac{(z+1)}{2}\partial_{\alpha_j}\big)F_z(\boldsymbol{\alpha}), \tag{3.12b}$$

$$\rho\hat{a}_j^\dagger \to \big(\alpha_j^* - \frac{(z-1)}{2}\partial_{\alpha_j}\big)F_z(\boldsymbol{\alpha}). \tag{3.12c}$$

From these basic correspondences, we can find the correspondence when acting with any number of annihilation or creation operators. Let us show this for the terms appearing in our master equation (2.40).

3.2.1 Evolution due to $\hat{H}_0$

The term corresponding to the free evolution of the cavity modes gives the correspondence (we use a single-mode description for simplicity, which can be applied to any of our modes):

$$-\mathrm{i}\big[\Delta\hat{a}^\dagger\hat{a},\rho\big] = -\mathrm{i}(\Delta\hat{a}^\dagger\hat{a}\rho - \rho\Delta\hat{a}^\dagger\hat{a}) \tag{3.13}$$
$$\downarrow$$
$$-\mathrm{i}\Delta\Big(\big(\alpha^* - \frac{(z+1)}{2}\partial_\alpha\big)\big(\alpha - \frac{(z-1)}{2}\partial_{\alpha^*}\big) - \big(\alpha - \frac{(z+1)}{2}\partial_{\alpha^*}\big)\big(\alpha^* - \frac{(z-1)}{2}\partial_\alpha\big)\Big)F_z(\alpha),$$

which once simplified reads as

$$\mathrm{i}\Delta(\partial_\alpha\alpha - \partial_{\alpha^*}\alpha^*)F_z(\alpha), \tag{3.14}$$

which is independent of z, so the dynamics of the distribution due to free evolution is the same irrespective of the representation.

3.2.1.1 Evolution due to $\hat{H}_{inj}$

The evolution of the state due to the laser drive provides the correspondence

$$\big[\epsilon\hat{a}^\dagger - \epsilon^*\hat{a},\rho\big] = \epsilon\hat{a}^\dagger\rho - \rho\epsilon\hat{a}^\dagger - \epsilon^*\hat{a}\rho + \rho\epsilon^*\hat{a} \tag{3.15}$$
$$\downarrow$$
$$\Big(\epsilon\big(\alpha^* - \frac{(z+1)}{2}\partial_\alpha\big) - \epsilon\big(\alpha^* - \frac{(z-1)}{2}\partial_\alpha\big) - \epsilon^*\big(\alpha - \frac{(z-1)}{2}\partial_{\alpha^*}\big) + \epsilon^*\big(\alpha - \frac{(z+1)}{2}\partial_{\alpha^*}\big)\Big)F_z(\alpha),$$

which is simplified as

$$-\big(\epsilon^*\partial_{\alpha^*} + \epsilon\partial_\alpha\big)F_z(\alpha), \tag{3.16}$$

which is again the same irrespective of the representation.

3.2.2 Evolution due to $\hat{H}_{NL}$

The evolution of the state due to the coherent part of parametric down-conversion provides the correspondence

$$\big[\epsilon_p\hat{a}_i^\dagger\hat{a}_s^\dagger - \epsilon_p^*\hat{a}_i\hat{a}_s,\rho\big] = \epsilon_p\hat{a}_i^\dagger\hat{a}_s^\dagger\rho - \epsilon_p^*\hat{a}_i\hat{a}_s\rho - \rho\epsilon_p\hat{a}_i^\dagger\hat{a}_s^\dagger + \rho\epsilon_p^*\hat{a}_i\hat{a}_s \tag{3.17}$$
$$\downarrow$$
$$\begin{aligned}\big(-\epsilon_p^*\big(\alpha_i - \frac{(z-1)}{2}\partial_{\alpha_i^*}\big)\big(\alpha_s - \frac{(z-1)}{2}\partial_{\alpha_s^*}\big) + \epsilon_p\big(\alpha_i^* - \frac{(z+1)}{2}\partial_{\alpha_i}\big)\big(\alpha_s^* - \frac{(z+1)}{2}\partial_{\alpha_s}\big)\\ +\epsilon_p^*\big(\alpha_i - \frac{(z+1)}{2}\partial_{\alpha_i^*}\big)\big(\alpha_s - \frac{(z+1)}{2}\partial_{\alpha_s^*}\big) - \epsilon_p\big(\alpha_i^* - \frac{(z-1)}{2}\partial_{\alpha_i}\big)\big(\alpha_s^* - \frac{(z-1)}{2}\partial_{\alpha_s}\big)\big)F_z(\boldsymbol{\alpha}),\end{aligned}$$

which after simplification leads to

$$\big(z\left(\epsilon_p^*\partial_{\alpha_i^*}\partial_{\alpha_s^*} + \epsilon_p\partial_{\alpha_i}\partial_{\alpha_s}\right) - \epsilon_p^*\alpha_i\partial_{\alpha_s^*} - \epsilon_p^*\alpha_s\partial_{\alpha_i^*} - \epsilon_p\alpha_i^*\partial_{\alpha_s} - \epsilon_p\alpha_s^*\partial_{\alpha_i}\big)F_z(\boldsymbol{\alpha}) \tag{3.18}$$

which in this case is clearly dependent on the choice of distribution. In particular, the second order derivatives do not contribute in the Wigner representation ($z = 0$).

3.2.3 Evolution due to single photon losses

The evolution due to single photon losses, provides the correspondence

$$\begin{aligned}
&\gamma(2\hat{a}\rho\hat{a}^\dagger - \hat{a}^\dagger\hat{a}\rho - \rho\hat{a}^\dagger\hat{a}) \qquad (3.19)\\
&\downarrow\\
&\gamma\Big(2(\alpha - \frac{z-1}{2}\partial_{\alpha^*})(\alpha^* - \frac{z-1}{2}\partial_\alpha) - (\alpha^* - \frac{z+1}{2}\partial_\alpha)(\alpha - \frac{z-1}{2}\partial_{\alpha^*})\\
&\qquad - (\alpha - \frac{z+1}{2}\partial_{\alpha^*})(\alpha^* - \frac{z-1}{2}\partial_\alpha)\Big)F_z(\alpha),
\end{aligned}$$

which simplifies as

$$\gamma\big(\partial_\alpha\alpha + (1-z)\partial_{\alpha^*}\partial_\alpha + \partial_{\alpha^*}\alpha^*\big). \qquad (3.20)$$

Again, this expression depends on the choice of distribution, and specifically it has only first order derivatives for the Glauber-Sudarshan representation ($z = 1$).

3.2.4 Evolution due to two-photon losses

The evolution due to two-photon losses provides the correspondence

$$\begin{aligned}
&\kappa(2\hat{a}_s\hat{a}_i\rho\hat{a}_i^\dagger\hat{a}_s^\dagger - \hat{a}_s^\dagger\hat{a}_i^\dagger\hat{a}_s\hat{a}_i\rho - \rho\hat{a}_s^\dagger\hat{a}_i^\dagger\hat{a}_s\hat{a}_i) \qquad (3.21)\\
&\downarrow\\
&\kappa\big(2(\alpha_s - \frac{z-1}{2}\partial_{\alpha_s^*})(\alpha_i - \frac{z-1}{2}\partial_{\alpha_i^*})(\alpha_s^* - \frac{z-1}{2}\partial_{\alpha_s})(\alpha_i^* - \frac{z-1}{2}\partial_{\alpha_i})\\
&-(\alpha_s^* - \frac{z+1}{2}\partial_{\alpha_s})(\alpha_i^* - \frac{z+1}{2}\partial_{\alpha_i})(\alpha_s - \frac{z-1}{2}\partial_{\alpha_s^*})(\alpha_i - \frac{z-1}{2}\partial_{\alpha_i^*})\\
&-(\alpha_s - \frac{z+1}{2}\partial_{\alpha_s^*})(\alpha_i - \frac{z+1}{2}\partial_{\alpha_i^*})(\alpha_s^* - \frac{z-1}{2}\partial_{\alpha_s})(\alpha_i^* - \frac{z-1}{2}\partial_{\alpha_i})\big)F_z(\boldsymbol{\alpha}),
\end{aligned}$$

which simplifying and collecting all the terms with the same order of derivatives leads to

$$\begin{aligned}
\kappa\Bigg[&\partial_{\alpha_s}\big(|\alpha_i|^2\alpha_s\big) + \partial_{\alpha_i}\big(|\alpha_s|^2\alpha_i\big) + \partial_{\alpha_s^*}\big(|\alpha_i|^2\alpha_s^*\big) + \partial_{\alpha_i^*}\big(|\alpha_s|^2\alpha_i^*\big) \qquad (3.22)\\
&+ \partial_{\alpha_s}\partial_{\alpha_i}\big(-z\alpha_i\alpha_s\big) + \partial_{\alpha_s^*}\partial_{\alpha_i^*}\big(-z\alpha_i^*\alpha_s^*\big) + \partial_{\alpha_s}\partial_{\alpha_i^*}\big((1-z)\alpha_i^*\alpha_s\big) + \partial_{\alpha_s^*}\partial_{\alpha_i}\big((1-z)\alpha_i\alpha_s^*\big)\\
&+ \partial_{\alpha_s}\partial_{\alpha_s^*}\big((1-z)\alpha_i\alpha_i^*\big) + \partial_{\alpha_i}\partial_{\alpha_i^*}\big((1-z)\alpha_s\alpha_s^*\big) - \partial_{\alpha_s}\partial_{\alpha_s}\partial_{\alpha_i^*}\partial_{\alpha_i}\frac{z}{2}\big(z-1\big)^2\\
&+ \partial_{\alpha_s}\partial_{\alpha_s^*}\partial_{\alpha_i^*}\frac{1}{4}\alpha_i^*\big(z-1\big)\big(3z-1\big) + \partial_{\alpha_s^*}\partial_{\alpha_i}\partial_{\alpha_i^*}\frac{1}{4}\alpha_s^*\big(z-1\big)\big(3z-1\big)\\
&+ \partial_{\alpha_s}\partial_{\alpha_s^*}\partial_{\alpha_i}\frac{1}{4}\alpha_i\big(z-1\big)\big(3z-1\big) + \partial_{\alpha_s}\partial_{\alpha_i^*}\partial_{\alpha_i}\frac{1}{4}\alpha_s\big(z-1\big)\big(3z-1\big)\Bigg]F_z(\boldsymbol{\alpha}).
\end{aligned}$$

In this case, all the distributions get terms with second order derivatives, but only the Glauber-Sudarshan distribution is free of third and fourth order derivatives.

3.2.5 Full evolution equation of the quasiprobability distribution

Collecting all the terms found above, we can finally write the general evolution equation for the quasiprobability distribution associated to our master equation (2.40):

$$
\begin{aligned}
\partial_t F_z(\boldsymbol{\alpha}) = \Bigg[& \partial_{\alpha_s}\big(\kappa|\alpha_i|^2\alpha_s - \epsilon_p\alpha_i^* + \gamma\alpha_s + \mathrm{i}\Delta\alpha_s - \epsilon_s\big) + \partial_{\alpha_i}\big(\kappa|\alpha_s|^2\alpha_i - \epsilon_p\alpha_s^* + \gamma\alpha_i + \mathrm{i}\Delta\alpha_i - \epsilon_i\big) \\
& + \partial_{\alpha_s^*}\big(\kappa|\alpha_i|^2\alpha_s^* - \epsilon_p^*\alpha_i + \gamma\alpha_s^* - \mathrm{i}\Delta\alpha_s^* - \epsilon_s^*\big) + \partial_{\alpha_i^*}\big(\kappa|\alpha_s|^2\alpha_i^* - \epsilon_p^*\alpha_s + \gamma\alpha_i^* - \mathrm{i}\Delta\alpha_i^* - \epsilon_i^*\big) \\
& + \partial_{\alpha_s}\partial_{\alpha_i} z\big(\epsilon_p - \kappa\alpha_i\alpha_s\big) + \partial_{\alpha_s^*}\partial_{\alpha_i^*} z\big(\epsilon_p^* - \kappa\alpha_i^*\alpha_s^*\big) + \partial_{\alpha_s}\partial_{\alpha_i^*}\big(-\kappa(z-1)\alpha_i^*\alpha_s\big) \\
& + \partial_{\alpha_s^*}\partial_{\alpha_i}\big(-\kappa(z-1)\alpha_i\alpha_s^*\big) + \partial_{\alpha_s}\partial_{\alpha_s^*}(1-z)\big(\gamma + \kappa\alpha_i\alpha_i^*\big) + \partial_{\alpha_i}\partial_{\alpha_i^*}(1-z)\big(\gamma + \kappa\alpha_s\alpha_s^*\big) \\
& + \partial_{\alpha_s}\partial_{\alpha_s^*}\partial_{\alpha_i^*}\frac{1}{4}\kappa\alpha_i^*(z-1)(3z-1) + \partial_{\alpha_s^*}\partial_{\alpha_i}\partial_{\alpha_i^*}\frac{1}{4}\kappa\alpha_s^*(z-1)(3z-1) \\
& + \partial_{\alpha_s}\partial_{\alpha_s^*}\partial_{\alpha_i}\frac{1}{4}\kappa\alpha_i(z-1)(3z-1) + \partial_{\alpha_s}\partial_{\alpha_i^*}\partial_{\alpha_i}\frac{1}{4}\kappa\alpha_s(z-1)(3z-1) \\
& - \partial_{\alpha_s}\partial_{\alpha_s}\partial_{\alpha_i^*}\partial_{\alpha_i}\frac{\kappa z}{2}(z-1)^2\Bigg] F_z(\boldsymbol{\alpha}).
\end{aligned}
\tag{3.23}
$$

By substituting the corresponding value of z we can get an expression for the different representations.

In spite of their more classical look as compared to the master equation, these partial differential equations are still not analytically solvable, and are even difficult to treat numerically. In the next chapters we discuss an alternate method that will allow us to find good approximations for expectation values of observables and for the distributions in an easier fashion.

CHAPTER 4

FOKKER-PLANCK AND STOCHASTIC LANGEVIN EQUATIONS

The partial differential equation derived in previous chapter is almost as difficult to treat as the original master equation. In this chapter, we explore its connection to well known tools from classical statistical mechanics. Specifically, we will introduce the Fokker-Planck equation for probability distributions, and show how it can be mapped to a set of stochastic ordinary differential equations. We will then show that, even though none of the quasiprobability distributions we have introduced satisfy a Fokker-Planck equation, a generalized form of the Glauber-Sudarshan (positive P representation) does satisfy it. Therefore, this technique will allow us to turn the master equation into a set of first order ordinary differential equations with noise that we will be able to treat more easily.

4.1 Classical master equation

Just like the master equation derived in chapter 3 for open quantum system, one can find a master equation for the time evolution of the probability density function corresponding to a stochastic Markovian process $x(t)$ (for simplicity, we consider a single variable, but everything is straghtforwardly geenralized to any number of variables), which in general reads

$$\partial_t P(x,t) = \int_{\mathbb{R}} dx' \big(W(x|x')P(x',t) - W(x'|x)P(x,t) \big), \tag{4.1}$$

being $W(x|x')$ the transition probability per unit time from x' to x. Let us define the 'distance' between x and x' as $r = x - x'$, and change slightly the notation for the transition probability as $W(x';r) = W(x' - r|x')$, so that it can be interpreted as the probability of departing from x' by a distance r. The master equation (4.1) then reads,

$$\partial_t P(x,t) = \int_{\mathbb{R}} dr W(x-r;r)P(x-r,t) - P(x,t)\int_{\mathbb{R}} dr W(x;-r) \tag{4.2}$$

By assuming that $W(x';r)$ is a sharply peaked around $r=0$ but varies slowly with x', and that $P(x,t)$ also varies slowly with x, we use Taylor expansion to get

$$\begin{aligned}\frac{\partial P(x,t)}{\partial t} &= \int_{\mathbb{R}} dr W(x;r)P(x,t) + \sum_{n=1}^{\infty}\frac{(-1)^n}{n!}\int_{\mathbb{R}} dr r^n \frac{\partial^n}{\partial x^n}\big(W(x;r)P(x,t)\big) \\ &- P(x,t)\int_{\mathbb{R}} dr W(x;-r) = \sum_{n=1}^{\infty}\frac{(-1)^n}{n!}\frac{\partial^n}{\partial x^n}\Big[\Big(\int_{\mathbb{R}} dr r^n W(x;r)\Big)P(x,t)\Big]\end{aligned} \tag{4.3}$$

where we have used $\int_{\mathbb{R}} dr W(x;-r) = \int_{\mathbb{R}} dr W(x;r)$ as trivially seen from the variable change $r \to -r$. This is known as Kramers-Moyal expansion of the master equation, which can be written as

$$\frac{\partial P(x,t)}{\partial t} = \sum_{n=1}^{\infty}\frac{(-1)^n}{n!}\frac{\partial^n}{\partial x^n}\big(K_n(x)P(x,t)\big), \tag{4.4}$$

so that all information about the stochastic dynamics is contained in the functions

$$K_n(x) = \int_{\mathbb{R}} dr r^n W(x;r). \tag{4.5}$$

4.2 Fokker-Planck equation

Many of the statistical processes encountered in classical mechanics, such as brownian motion, can be modelled by just the first two terms of the Kramers-Moyal expansion:

$$\partial_t P(x,t) = \Big(-\partial_x A(x) + \frac{1}{2}\partial_x^2 D(x)\Big)P(x,t), \tag{4.6}$$

where $D(x) \geq 0$ as evidenced by (4.5). This is known as the Fokker-Planck equation and has been studied extensively in classical mechanics.

We can give some meaning to the $A(x)$ and $D(x)$ functions. For this, we consider the time evolution of the mean and the variance of the stochastic process. In the case of the mean, we have

$$\frac{d\langle x\rangle}{dt} = \frac{d}{dt}\int_{-\infty}^{\infty} dx\, xP(x,t) = \int_{-\infty}^{\infty} dx\, x\partial_t P(x,t). \tag{4.7}$$

Using the Fokker-Planck equation (4.21) and integrating it by parts (assuming the distribution and its derivative vanish sufficiently fast at infinity) we find

$$\frac{d\langle x\rangle}{dt} = \langle A(x)\rangle \tag{4.8}$$

Similarly we can find the evolution of the variance $\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2$ as

$$\frac{d\sigma^2}{dt} = \frac{d}{dt}\langle x^2 \rangle - 2\langle x \rangle \frac{d\langle x \rangle}{dt} = 2\langle xA(x) \rangle - 2\langle x \rangle \langle A(x) \rangle + \langle D(x) \rangle. \tag{4.9}$$

$A(x)$ is then called *drift* term as it imparts drift to the distribution. If there is no drift, the evolution of the variance is governed by $D(x)$ alone. Hence $D(x)$ is called *diffusion* as it accounts for the dynamics of the fluctuations.

4.3 Langevin approach

There is an alternative approach to describe the evolution of the stochastic process $x(t)$. In particular, instead of using an equation of motion for its underlaying probability distribution $P(x,t)$, we can use an evolution equation for the process $x(t)$ itself, subject to some form of noise (this is actually similar in spirit to the distinction between the Schrödinger and Heisenberg pictures in quantum mechanics). This way we would obtain a stochastic differential equation which receives the name of stochastic Langevin equation. In this section we discuss the stochastic equation that corresponds to a Fokker-Planck equation.

Since we have an interpretation for the terms that define the Fokker-Planck equation, namely $A(x)$ and $D(x)$, we already have a clue as to how the corresponding Langevin equation might look. Specifically, we expect it to have a deterministic part that will rule the evolution of the mean, and will be therefore given by the drift $A(x)$, that is,

$$\frac{dx}{dt} = A(x) + \text{stochastic part}. \tag{4.10}$$

The question now is what is the form of the stochastic term. Since we are dealing with Markovian processes, let us write it in terms of a Gaussian white noise $\eta(t)$, with statistical properties

$$\langle \eta(t) \rangle = 0, \qquad \langle \eta(t)\eta(t') \rangle = \delta(t-t'). \tag{4.11}$$

Higher order correlation functions can be obtained form the above correlation, since the noise is assumed to be Gaussian. Given this noise, we then make the following ansatz for the Langevin equation:

$$\frac{dx}{dt} = A(x) + B(x)\eta(t), \tag{4.12}$$

so all that is left is to find $B(x)$, which we will call *noise* term. In order to do this, we will evaluate the variance of the stochastic process, and compare it with what we found in the previous section from the Fokker-Planck equation. For simplicity, we further take $A(x) = 0$ and assume that $B(x)$ is independent of x. The solution of the Langevin equation is

$$x(t) = x(0) + B\int_0^t dt' \eta(t), \tag{4.13}$$

which shows that the mean does not evolve, $\langle x(t)\rangle = \langle x(0)\rangle$, as expected. On the other hand, we have

$$\langle x^2(t)\rangle = \langle x^2(0)\rangle + B^2 \int_0^t dt' \int_0^t dt'' \underbrace{\langle \eta(t')\eta(t'')\rangle}_{\delta(t'-t'')} = \langle x^2(0)\rangle + B^2 t, \tag{4.14}$$

leading to a variance $\sigma^2(t) = \sigma^2(0) + B^2 t$. Therefore, we find that the evolution equation of the variance is

$$\frac{d\sigma^2}{dt} = B^2. \tag{4.15}$$

By comparing this with the result obtained with the Fokker-Planck equation (4.21), we find that $B = \sqrt{D}$. In order to extend this relation to a general diffusion $D(x)$, we need to discuss some subtleties of stochastic calculus.

4.4 Wiener process and stochastic calculus

Technically speaking, equation (4.12) is not well defined as a differential equation, since $\eta(t)$ is not continuous. In the following we will introduce a slightly different formulation that leads to a well defined form of stochastic calculus, and will allow us to prove the relation $B(x) = \sqrt{D(x)}$ in general.

The formulation is based on understanding the differential equation as a continuous limit of some discretized version where the time t takes values only in the discrete interval $\{t_k = k\Delta t\}_{k=0,1,\ldots}$, finally taking the limit $\Delta t \to 0$. In this context, one defines the so-called Wiener increment as

$$dW(t) = \int_t^{t+\Delta t} dt' \eta(t'). \tag{4.16}$$

which is a Gaussian random variable with correlation functions that are well defined as a function of the time step Δt, specifically:

$$\langle dW(t)\rangle = 0, \qquad \langle dW(t_k) dW(t_l)\rangle = \delta_{lk}\Delta t \tag{4.17}$$

One then writes the discretized form of the Langevin equation as

$$dx = A(x)\Delta t + B(x) dW(t) \tag{4.18}$$

where $dx = x(t+\Delta t) - x(t)$, which has no problems of definition, and therefore provides meaning to the continuous form of the Langevin equation in the limit $\Delta t \to 0$ (note that it might seem that we are deliberately mixing infinitesimal and discrete increments in the notation, but it all makes sense in the $\Delta t \to 0$ limit that is implicitly understood in the following).

The Wiener increment allows us to make a consistent definition of integrals of stochastic functions. However, as we will see in a second, such definition is not unique, and it leads to different interpretations of the Langevin equation and different forms of stochastic calculus. This can be seen when considering integrals of the type $I(t) = \int_0^t dW(t')g\big(W(t')\big)$, where g is an arbitrary function. When defining this integral as the continuous limit of a Riemman sum,

$$I_\alpha(t_n) = \int_0^{t_n} dW(t')g\big(W(t')\big) \approx \sum_{k=0}^{n-1}[W(t_{k+1}) - W(t_k)]g\big(W(\tau_k)\big), \tag{4.19}$$

where $\tau_k = t_k + \alpha\Delta t$ with $\alpha \in [0,1]$, we are free to choose where to evaluate the function g within the discrete time intervals, that is, we are free to choose the value of α. For the deterministic processes we are used to, the results are independent of α. However, for stochastic processes, the results can depend very strongly on the choice of α. Let us consider the simplest example, where $g\big(W(t)\big) = W(t)$. For this case, it can be shown that the integral takes the expentation value $\langle I_\alpha(t_n)\rangle = \alpha t_n$, which is linear in α. Therefore, for a given Langevin equation, different choices of α give rise to different physical results, which means that one has to be careful to choose the right α for the given physical situation.

In the case of the connection between Langevin and Fokker-Planck equations, all the derivations that we did in the previous section for constant difusion D and noise term B would apply to the general case if the Wiener increment and $B(x)$ were statistically independent. It is possible to show [20, 21] that this is exactly what happens with the choice $\alpha = 0$, which gives rise to the so called Ito interpretation of the Langevin equation. Therefore, we can made the connection $B(x) = \sqrt{D(x)}$ between Langevin and Fokker-Planck equations, only when using the Ito interpretation. The price to pay is that when setting $\alpha = 0$ the usual rules of calculus (chain rule, product rule, etc) do not apply anymore. This is only the case when choosing $\alpha = 1/2$, which leads to the so-called Stratonovich interpretation of the Langevin equation [20, 21]. Thankfully, given an Ito Langevin equation, it is always possible to transform it into a Stratonovich Langevin equation simply by changing the drift term as [20, 21]

$$A'(x) = A(x) + \frac{1}{2}B(x)\partial_x B(x). \tag{4.20}$$

Hence, when the noise term is independent of the stochastic process x, both interpretations coincide. This will indeed be the case within the so-called linearized approximation that we will apply to our system in order to find analytical results.

4.5 Fokker-Planck and Langevin equations in higher dimensions

For our future purposes, we will need to go beyond the single-variable case exposed in the previous sections [20, 21]. Consider then N stochastic processes collected in the vector

$\mathbf{x} = (x_1, x_2, ..., x_N)^{\mathrm{T}}$. The general form of the Fokker-Planck equation is now

$$\partial_t P(\mathbf{x}, t) = \left(-\sum_{j=1}^{N} \partial_j A_j(\mathbf{x}) + \frac{1}{2} \sum_{j,l=1}^{N} \partial_j \partial_l D_{jl}(\mathbf{x}) \right) P(\mathbf{x}, t) \tag{4.21}$$

where the drift and diffusion terms can now be collected in the so-called *drift vector* $\mathbf{A}(\mathbf{x})$ and *diffusion matrix* $D(\mathbf{x})$. The requirement for the probability distribution to remain physical is now that the diffusion matrix must remain positive semi-definite, $D(\mathbf{x}) \geq 0$, that is, is not allowed to have negative eigenvalues. Note that we are further simplifying the notation for partial derivatives with respect to the stochastic variables as $\partial_j = \partial/\partial x_j$ when required.

The equivalent set of stochastic Langevin equations takes now the form

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(\mathbf{x}) + B(\mathbf{x})\boldsymbol{\eta}(t), \tag{4.22}$$

where the *noise matrix* $B(\mathbf{x})$ is related to the diffusion matrix by $D(x) = B(\mathbf{x})B^{\mathrm{T}}(\mathbf{x})$, and the elements of the vector $\boldsymbol{\eta}(t)$ are statistically-independent real Gaussian white noises which satisfy

$$\langle \eta_j(t) \rangle = 0, \qquad \langle \eta_j(t)\eta_l(t') \rangle = \delta_{jl}\delta(t - t'). \tag{4.23}$$

4.6 Connection between the Fokker-Planck equation and our phase-space dynamics

With these tools from classical statistical mechanics at hand, we can now think about applying them to the equation of motion (3.23) satisfied by our phase-space quasiprobability distributions $F_z(\boldsymbol{\alpha})$. The first objection one can raise is that these are not even true probability distributions in the classical sense (except for $z = -1$). It's clear from (3.23) that neither the Wigner nor the Husimi distribution obey a Fokker-Planck equation, since their evolution equation contains derivatives beyond second order in the phase-space variables. In contrast, the equation for the Glauber-Sudarshan distribution ($z = 1$) looks like Fokker-Planck equation. We then need to check if the corresponding diffusion matrix is positive semidefinite. Since the Fokker-Planck equation is defined for real variables, we first need to change to the real phase-space variables (position and momentum). This is easily achieved by using the following expressions

$$x_j = \frac{1}{2}(\alpha_j + \alpha_j^*), \qquad p_j = -\frac{\mathrm{i}}{2}(\alpha_j - \alpha_j^*), \tag{4.24a}$$

$$\alpha_j = (x_j + \mathrm{i}p_j), \qquad \alpha^* = (x_j - \mathrm{i}p_j), \tag{4.24b}$$

$$\partial_{\alpha_j} = \partial_{x_j} - \mathrm{i}\partial_{p_j}, \qquad \partial_{\alpha_j^*} = \partial_{x_j} + \mathrm{i}\partial_{p_j}. \tag{4.24c}$$

For $z=1$, we can then rewrite the term of (3.23) with second-order derivatives as

$$\begin{aligned}
&\partial_{\alpha_s}\partial_{\alpha_i}(-\kappa\alpha_i\alpha_s+\epsilon_p)+\partial_{\alpha_s^*}\partial_{\alpha_i^*}(-\kappa\alpha_s^*\alpha_i^*+\epsilon_p^*) \\
&\qquad\qquad = (\partial_{p_s}\partial_{p_i}-\partial_{x_s}\partial_{x_i})\underbrace{\big(\kappa(x_ix_s-p_ip_s)+2\mathrm{Re}\{\epsilon_p\}\big)}_{d_1} \\
&\qquad\qquad + (\partial_{p_s}\partial_{x_i}-\partial_{x_s}\partial_{p_i})\underbrace{\big(\kappa(x_ip_s-p_ix_s)+2\mathrm{Im}\{\epsilon_p\}\big)}_{d_2}
\end{aligned} \tag{4.25}$$

Choosing the order (x_s,p_s,x_i,p_i) for the phase-space variables, the diffusion matrix reads

$$\begin{pmatrix} 0 & 0 & -d_1 & -d_2 \\ 0 & 0 & d_2 & d_1 \\ -d_1 & d_2 & 0 & 0 \\ -d_2 & d_1 & 0 & 0 \end{pmatrix}, \tag{4.26}$$

with eigenvalues $\pm(d_1+d_2)$ and $\pm(d_1-d_2)$. Hence, the diffusion is not positive semidefinite, and not even for the Glauber-Sudarshan distribution we get a true Fokker-Planck equation.

Despite this apparent disappointment, in the next section we show how to generalize this distribution such that we obtain a true Fokker-Planck equation.

4.7 Positve P-representation and stochastic Langevin equations of our system

Let's proceed for a second as if the diffusion matrix studied above would be positive semidefinite, in order to understand where we would find a problem in defining a set of associated Langevin equations. Staying in the complex representation for simplicity and choosing the order $\boldsymbol{\alpha}=(\alpha_s,\alpha_i,\alpha_s^*,\alpha_i^*)^{\mathrm{T}}$, the diffusion matrix takes the form

$$D(\boldsymbol{\alpha})=\begin{pmatrix} 0 & \beta & 0 & 0 \\ \beta & 0 & 0 & 0 \\ 0 & 0 & 0 & \beta^* \\ 0 & 0 & \beta^* & 0 \end{pmatrix}, \tag{4.27}$$

where we have defined $\beta=\epsilon_p-\kappa\alpha_i\alpha_s$, which can be decomposed as $D=BB^{\mathrm{T}}$ with

$$B(\boldsymbol{\alpha})=\begin{pmatrix} \sqrt{\beta} & \mathrm{i}\sqrt{\beta} & 0 & 0 \\ \sqrt{\beta} & -\mathrm{i}\sqrt{\beta} & 0 & 0 \\ 0 & 0 & \sqrt{\beta^*} & \mathrm{i}\sqrt{\beta^*} \\ 0 & 0 & \sqrt{\beta^*} & -\mathrm{i}\sqrt{\beta^*} \end{pmatrix}, \tag{4.28}$$

leading to Langevin equations with the following noise terms:

$$\dot{\alpha}_s = ... + \sqrt{\beta}[\eta_1(t) + \mathrm{i}\eta_2(t)], \tag{4.29a}$$
$$\dot{\alpha}_i = ... + \sqrt{\beta}[\eta_1(t) - \mathrm{i}\eta_2(t)], \tag{4.29b}$$
$$\dot{\alpha}_s^* = ... + \sqrt{\beta^*}[\eta_3(t) + \mathrm{i}\eta_4(t)], \tag{4.29c}$$
$$\dot{\alpha}_i^* = ... + \sqrt{\beta^*}[\eta_3(t) - \mathrm{i}\eta_4(t)], \tag{4.29d}$$

where the noises $\eta_j(t)$ are independent and real. We can now understand the issue that arises when trying to build a set of Langevin equations: the noise terms in α_j and α_j^* are not complex-conjugate, which means that during the evolution, the equations do not keep the complex-conjugacy of these variables, and therefore we abandon the physical phase-space of the modes.

A more positive way of looking at this issue is as follows: it seems that if we extend the phase space of the modes to a four-dimensional one where we can take α_j and α_j^* as independent variables that do not need to be complex-conjugate, we will indeed be able to describe the quantum dynamics by a set of classical Langevin equations. This is precisely what the positive P representation does [22, 23]. Let us introduce it now in detail. Like in the previous chapter, we start by defining its characteristic function (we use a single-mode description again to simplify the notation)

$$\chi^+(\beta, \beta^+) = \mathrm{tr}\{\rho e^{\beta\hat{a}^\dagger} e^{-\beta^+\hat{a}}\}. \tag{4.30}$$

This function depends on two complex variables β and β^+ which are fully independent, in particular, $\beta^* \neq \beta^+$. Our inclination at this point would be to take the four-dimensional Fourier transform of the characteristic function in order to define the corresponding probability distribution. However, this would mean introducing explicit dependence on α^* and $(\alpha^+)^*$ that would not lead us anywhere useful. Instead, we simply generalize the relation (3.4) that we established in the previous chapter as:

$$\chi^+(\beta, \beta^+) = \int_{\mathbb{C}^2} d^2\alpha d^2\alpha^+ e^{\beta\alpha^+ - \beta^+\alpha} P^+(\alpha, \alpha^+), \tag{4.31}$$

which ensures that the positive P distribution $P^+(\alpha, \alpha^+)$ is normalized as a function of the two independent complex variables α and α^+, that is,

$$\int_{\mathbb{C}^2} d^2\alpha d^2\alpha^+ P^+(\alpha, \alpha^+) = 1. \tag{4.32}$$

In addition, it also ensures that the distribution provides normally ordered quantum expectation values as

$$\langle \hat{a}^{\dagger m}\hat{a}^n\rangle = (-1)^n \partial_\beta^m \partial_{\beta^+}^n \chi^+(\beta, \beta^+)\big|_{\beta=\beta^+=0} = \int_{\mathbb{C}^2} d^2\alpha d^2\alpha^+ P^+(\alpha, \alpha^+)\alpha^{+m}\alpha^n. \tag{4.33}$$

However, by not taking the four-dimensional Fourier transform, we loose the one to one correspondence between the distribution P^+ and the characteristic function χ^+, meaning that the distribution is not uniquely defined for a given quantum state $\hat{\rho}$. Nevertheless, it is exactly this lack of uniqueness that later on we will exploit in order to ensure that the distribution satisfies a true Fokker-Planck equation in the extended four-dimensional phase space defined by (α, α^+).

It is also interesting to mention that a direct connection between the positive P distribution and the state can be found, and is given by [22, 23]:

$$\hat{\rho} = \int_{\mathbb{C}^2} d^2\alpha d^2\alpha^+ P^+(\alpha, \alpha^+) \frac{|\alpha\rangle\langle\alpha^{+*}|}{\langle\alpha|\alpha^{+*}\rangle} \tag{4.34}$$

Note that this means that the distribution $P(\alpha, \alpha^+) = \delta(\alpha - \alpha_0)\delta(\alpha^+ - \alpha_0^*)$ corresponds to a coherent state $|\alpha_0\rangle$. In other words, choosing α and α^+ as deterministic and complex-conjugate variables, is equivalent to sampling a coherent state, a property that will be exploited later to connect with the classical limit.

One final interesting fact about the distribution is that, given a state $\hat{\rho}$, it can be chosen as [22, 23]

$$P^+(\alpha, \alpha^+) = \frac{1}{4\pi} e^{-|\alpha^* - \alpha^+|^2/4} \left\langle \frac{\alpha^* + \alpha^+}{2} \middle| \hat{\rho} \middle| \frac{\alpha^* + \alpha^+}{2} \right\rangle, \tag{4.35}$$

which shows that this distributions is real and positive.

We can find the dynamics of this distribution for our system in a similar way as discussed in the previous chapter for the other distributions. Specifically, it is straightforward to generalize the relations (3.11) and (3.12) as

$$\hat{a}_j\hat{\rho} \to \alpha_j P^+(\boldsymbol{\alpha}) \tag{4.36a}$$

$$\rho\hat{a}_j \to \left(\alpha_j - \partial_{\alpha_j^+}\right) P^+(\boldsymbol{\alpha}) \tag{4.36b}$$

$$\hat{a}_j^\dagger\rho \to \left(\alpha_j^+ - \partial_{\alpha_j}\right) P^+(\boldsymbol{\alpha}) \tag{4.36c}$$

$$\rho\hat{a}_j^\dagger \to \alpha_j^+ P^+(\boldsymbol{\alpha}), \tag{4.36d}$$

where we have collected the complex variables of the extended phase space as $\boldsymbol{\alpha} = (\alpha_s, \alpha_i, \alpha_s^+, \alpha_i^+)$. Then, given our master equation (2.40), the resulting partial differential equation for $P^+(\boldsymbol{\alpha})$ has the same form as that for the Glauber-Sudarshan one, that is, (3.23) with $z = 1$, but replacing α^* by α^+:

$$\begin{aligned}
\partial_t P^+(\boldsymbol{\alpha}) = \big(&\partial_{\alpha_s}\left((\gamma + \mathrm{i}\Delta_s + \kappa\alpha_i^+\alpha_i)\alpha_s - \epsilon_s - \epsilon_p\alpha_i^+\right) + \partial_{\alpha_i}\left((\gamma + \mathrm{i}\Delta_i + \kappa\alpha_s^+\alpha_s)\alpha_i - \epsilon_i - \epsilon_p\alpha_s^+\right) \\
&+ \partial_{\alpha_s^+}\left((\gamma - \mathrm{i}\Delta_s + \kappa\alpha_i^+\alpha_i)\alpha_s^+ - \epsilon_s^* - \epsilon_p^*\alpha_i\right) \\
&+ \partial_{\alpha_i^+}\left((\gamma - \mathrm{i}\Delta_i + \kappa\alpha_s^+\alpha_s)\alpha_i^+ - \epsilon_i^* - \epsilon_p^*\alpha_s\right) \\
&+ \partial_{\alpha_i}\partial_{\alpha_s}(-\kappa\alpha_i\alpha_s + \epsilon_p) + \partial_{\alpha_i^+}\partial_{\alpha_s^+}(-\kappa\alpha_i^+\alpha_s^+ + \epsilon_p^*)\big) P^+(\boldsymbol{\alpha}),
\end{aligned} \tag{4.37}$$

which corresponds to a Fokker-Planck equation with drift vector

$$\mathbf{A}(\boldsymbol{\alpha}) = \begin{pmatrix} \epsilon_s - (\gamma + \mathrm{i}\Delta_s + \kappa\alpha_i^+\alpha_i)\alpha_s + \epsilon_p\alpha_i^+ \\ \epsilon_i - (\gamma + \mathrm{i}\Delta_i + \kappa\alpha_s^+\alpha_s)\alpha_i + \epsilon_p\alpha_s^+ \\ \epsilon_s^* - (\gamma - \mathrm{i}\Delta_s + \kappa\alpha_i^+\alpha_i)\alpha_s^+ + \epsilon_p^*\alpha_i \\ \epsilon_i^* - (\gamma - \mathrm{i}\Delta_i + \kappa\alpha_s^+\alpha_s)\alpha_i^+ + \epsilon_p^*\alpha_s \end{pmatrix}, \tag{4.38}$$

and diffusion matrix

$$D(\boldsymbol{\alpha}) = \begin{pmatrix} 0 & \epsilon_p - \kappa\alpha_i\alpha_s & 0 & 0 \\ \epsilon_p - \kappa\alpha_i\alpha_s & 0 & 0 & 0 \\ 0 & 0 & 0 & \epsilon_p^* - \kappa\alpha_i^+\alpha_s^+ \\ 0 & 0 & \epsilon_p^* - \kappa\alpha_i^+\alpha_s^+ & 0 \end{pmatrix}. \tag{4.39}$$

Note that we keep a complex representation to simplify the notation. The key point now is that in both the distribution and the equation, there is no explicit dependence on the complex-conjugate variables $\boldsymbol{\alpha}^*$, so that all the functions are analytical in the complex sense. This can be exploited to prove that $P^+(\boldsymbol{\alpha})$ can be chosen such that the diffusion matrix is positive semidefinite for all $\boldsymbol{\alpha}$, and therefore, this is a true Fokker-Planck equation. The proof can be seen in [18, 22, 23].

We can then map the above equation to a set of stochastic Langevin equations. For this, we first need to decompose the diffusion matrix as $D(\boldsymbol{\alpha}) = B(\boldsymbol{\alpha})B^{\mathrm{T}}(\boldsymbol{\alpha})$. This is exactly what we did in (4.28), so we can write the noise matrix as

$$B(\boldsymbol{\alpha}) = \frac{1}{\sqrt{2}}\begin{pmatrix} \sqrt{\epsilon_p - \kappa\alpha_i\alpha_s} & \mathrm{i}\sqrt{\epsilon_p - \kappa\alpha_i\alpha_s} & 0 & 0 \\ \sqrt{\epsilon_p - \kappa\alpha_i\alpha_s} & -\mathrm{i}\sqrt{\epsilon_p - \kappa\alpha_i\alpha_s} & 0 & 0 \\ 0 & 0 & \sqrt{\epsilon_p^* - \kappa\alpha_i^+\alpha_s^+} & \mathrm{i}\sqrt{\epsilon_p^* - \kappa\alpha_i^+\alpha_s^+} \\ 0 & 0 & \sqrt{\epsilon_p^* - \kappa\alpha_i^+\alpha_s^+} & -\mathrm{i}\sqrt{\epsilon_p^* - \kappa\alpha_i^+\alpha_s^+} \end{pmatrix}, \tag{4.40}$$

leading to the stochastic Langevin equations

$$\dot{\boldsymbol{\alpha}} = \mathbf{A}(\boldsymbol{\alpha}) + B(\boldsymbol{\alpha})\boldsymbol{\eta}(t), \tag{4.41}$$

where $\boldsymbol{\eta}(t) = \big(\eta_1(t), \eta_2(t), \eta_3(t), \eta_4(t)\big)^{\mathrm{T}}$ has real independent white Gaussian noises as components. Then, we can write the stochastic equations for our system as

$$\dot{\alpha_s} = \epsilon_s - (\gamma + \mathrm{i}\Delta_s + \kappa\alpha_i^+\alpha_i)\alpha_s + \epsilon_p\alpha_i^+ + \sqrt{\epsilon_p - \kappa\alpha_i\alpha_s}\,\xi(t) \tag{4.42a}$$

$$\dot{\alpha_i} = \epsilon_i - (\gamma + \mathrm{i}\Delta_i + \kappa\alpha_s^+\alpha_s)\alpha_i + \epsilon_p\alpha_s^+ + \sqrt{(\epsilon_p - \kappa\alpha_i\alpha_s)}\,\xi^*(t) \tag{4.42b}$$

$$\dot{\alpha_s^+} = \epsilon_s^* - (\gamma - \mathrm{i}\Delta_s + \kappa\alpha_i^+\alpha_i)\alpha_s^+ + \epsilon_p^*\alpha_i + \sqrt{(\epsilon_p^* - \kappa\alpha_i^+\alpha_s^+)}\,\xi^+(t) \tag{4.42c}$$

$$\dot{\alpha_i^+} = \epsilon_i^* - (\gamma - \mathrm{i}\Delta_i + \kappa\alpha_s^+\alpha_s)\alpha_i^+ + \epsilon_p^*\alpha_s + \sqrt{(\epsilon_p^* - \kappa\alpha_i^+\alpha_s^+)}\,\big(\xi^+(t)\big)^* \tag{4.42d}$$

where $\xi(t) = \big(\eta_1(t) + \mathrm{i}\eta_2(t)\big)/\sqrt{2}$ and $\xi^+(t) = \big(\eta_3(t) + \mathrm{i}\eta_4(t)\big)/\sqrt{2}$ are independent complex white Gaussian noises with zero mean and only non-zero two-time correlation function

$$\langle \xi(t)\xi^*(t')\rangle = \langle \xi^+(t)\big(\xi^+(t)\big)^*\rangle = \delta(t-t'). \tag{4.43}$$

It is interesting to note that if we set the noise to zero, and replace α_j^+ by α_j^*, we retrieve the equations found in the classical limit of the system (2.45), just as expected from the connection between the positive P representation and coherent states commented above.

In order to simplify the theoretical analysis and get analytic insight, we are going to work in the so-called *symmetric configuration* [9], in which signal and idler are equally pumped ($|\epsilon_s| = |\epsilon_i|$) and have opposite detunings $\Delta_s = -\Delta_i > 0$. Furthermore, we take the pump phase as a reference ($\epsilon_p \in \mathbb{R} \geq 0$), and assume that the injection at the degenerate down-conversion frequency is linearly polarized and in phase with the pump ($\epsilon_s = \epsilon_i \in \mathbb{R} \geq 0$), what is known as the amplification regime. In order to reduce the number of free parameters in the system, it is then convenient to define the dimensionless time and parameters

$$\tau = \gamma t, \qquad g = \frac{\kappa}{\gamma}, \qquad \tilde{\Delta} = \frac{\Delta_s}{\gamma}, \qquad \mathcal{P} = \frac{1}{g}\left(\frac{\epsilon_s}{\gamma}\right)^2, \qquad \varepsilon = \frac{\epsilon_p}{\gamma}. \tag{4.44}$$

and the normalized variables and noises

$$\beta_j = \sqrt{g}\,\alpha_j, \qquad \beta_j^+ = \sqrt{g}\,\alpha_j^+, \qquad \eta(\tau) = \xi(\tau/\gamma)/\sqrt{\gamma}, \qquad \eta^+(\tau) = \xi^+(\tau/\gamma)/\sqrt{\gamma}. \tag{4.45}$$

The normalized Langevin equations now read

$$\dot{\beta}_s = \sqrt{\mathcal{P}} - (1+\mathrm{i}\tilde{\Delta})\beta_s + (\varepsilon - \beta_i\beta_s)\beta_i^+ + \sqrt{g(\varepsilon - \beta_i\beta_s)}\,\eta(\tau), \tag{4.46a}$$
$$\dot{\beta}_i = \sqrt{\mathcal{P}} - (1-\mathrm{i}\tilde{\Delta})\beta_i + (\varepsilon - \beta_i\beta_s)\beta_s^+ + \sqrt{g(\varepsilon - \beta_i\beta_s)}\,\eta^*(\tau), \tag{4.46b}$$
$$\dot{\beta}_s^+ = \sqrt{\mathcal{P}} - (1-\mathrm{i}\tilde{\Delta})\beta_s^+ + (\varepsilon - \beta_s^+\beta_i^+)\beta_i + \sqrt{g(\varepsilon - \beta_i^+\beta_s^+)}\,\eta^+(\tau), \tag{4.46c}$$
$$\dot{\beta}_i^+ = \sqrt{\mathcal{P}} - (1+\mathrm{i}\tilde{\Delta})\beta_i^+ + (\varepsilon - \beta_i^+\beta_s^+)\beta_s + \sqrt{g(\varepsilon - \beta_i^+\beta_s^+)}\,[\eta^+(\tau)]^*, \tag{4.46d}$$

where the dot denotes now derivative with respect to the dimensionless time τ, and the new noises satisfy the same statistical properties as the previous ones (independent, complex, white, and Gaussian) but with respect to the dimensionless time.

This normalization of variables and parameters is also useful in order to identify the classical limit of the master equation. Specifically, now we clearly see that the deterministic part dominates the evolution of the system only when $g \to 0$, limit in which quantum noise plays no role.

CHAPTER 5

CLASSICAL DYNAMICS AND LIMIT CYCLES

Now that we have a relatively simple set of first order differential equations for the dynamics of our quantum system, we can move on to analyzing its state. Since we are working with a dissipative system, we can expect it to relax to some sort of steady state, where the system no longer evolves indefinitely, but converges asymptotically in time to a specific solution irrespective of the initial state. In this chapter we analyze these states in the classical limit, which will be useful to tackle the quantum dynamics in the next chapters.

We have different types of asymptotic solutions depending on the choice for the parameters in the equations. It is common to start by finding the regions in the parameter space where the asymptotic solutions are stationary, and use linear stability analysis to study their stability. We will then find points in the parameter space where such solutions stop being stable, known as bifurcations. Using these bifurcation points as a reference we find regions in the parameter space where time-periodic asymptotic solutions known as limit cycles exist, which we find numerically.

5.1 Stationary solutions and linear stability analysis

Let us here introduce the general procedure that we will use to determine the asymptotic classical states of our system. Consider a system described by a set of variables $\mathbf{x} = (x_1, x_2, ..., x_N)^{\mathrm{T}}$, which evolve according to some set of coupled nonlinear first-order differential equations

$$\dot{\mathbf{x}} = \mathbf{A}(\mathbf{x}). \tag{5.1}$$

As mentioned above, when the system is dissipative, we can expect stationary solutions with $\dot{\mathrm{x}} = 0$ to exist as asymptotic solutions, that is,

$$\lim_{t\to\infty} \mathrm{x}(t) = \bar{\mathrm{x}}, \quad \text{where } \bar{\mathrm{x}} \text{ is found from the algebraic equations } \mathbf{A}(\bar{\mathrm{x}}) = \mathbf{0}. \tag{5.2}$$

For a given choice of the parameters of the equations, such stationary solution might be unique, or several might coexist, since the algebraic equations are nonlinear.

Once the stationary solutions are identified, one must check their stability against perturbations, since only when they are stable they will be accessible in real physical systems. Next we introduce the so-called linear stability analysis, which is the cornerstone for such analysis. A solution is said to be stable if any small perturbation is eventually washed off and the system comes back to its stationary state without external assistance. Therefore, the stability of a given stationary solution $\bar{\mathrm{x}}$ can be checked by considering small perturbations $\delta\mathrm{x}(t)$ around it, and studying their evolution. Let us then take $\mathrm{x}(t) = \bar{\mathrm{x}} + \delta\mathrm{x}(t)$ in (5.1), and use a first-order Taylor series expansion in the perturbations to write

$$\dot{\bar{x}}_j + \delta\dot{x}_j = A_j(\bar{\mathrm{x}}) + \sum_{l=1}^{N} \left.\frac{\partial A_j}{\partial x_l}\right|_{\mathrm{x}=\bar{\mathrm{x}}} \delta x_l. \tag{5.3}$$

Now, using the fact that $\bar{\mathrm{x}}$ is a stationary solution, so $\dot{\bar{x}}_j = 0 = A_j(\bar{\mathrm{x}})$, we can recast this equations into the linear system

$$\delta\dot{\mathrm{x}} = \mathcal{L}\,\delta\mathrm{x}, \tag{5.4}$$

where $\mathcal{L}$ is known as the linear stability matrix, with elements

$$\mathcal{L}_{jl} = \left.\frac{\partial A_j}{\partial x_l}\right|_{\mathrm{x}=\bar{\mathrm{x}}}. \tag{5.5}$$

Since (5.4) is linear, the eigenvalues of this matrix fully determine the fate of the perturbations $\delta\mathrm{x}$. Specifically, the perturbations will decay with time only when all the eigenvalues have negative real part, in which case we say that the system is stable. If at least one eigenvalue has positive real part, then the perturbations will grow in the direction determined by the corresponding eigenvector, and the solution will be unstable. The points connecting stable and unstable regions are called *bifurcations*, and are signaled by one eigenvalue that becomes either zero or purely imaginary. In the former case, we talk about a Pitchfork bifurcation, which usually means that our unstable stationary solution will depart towards another stationary solution. In the latter case, we talk about Hopf bifurcations, which usually mean that the unstable stationary solution departs towards some time-dependent asymptotic solution, which is typically time periodic and is known as limit cycle. In this way, even if limit cycles are not analytically accessible in most equations, we can guess the regions where we might find them from the bifurcations present in the stationary solutions.

5.2 Classical asymptotic solutions of our system

Let us apply now the general procedure explained above to equations (2.45) that describe our system in the classical limit. As explained at the end of the previous chapter, we will consider the symmetric configuration, that will allow us to find analytic results, as shown in [9]. In this regime, the classical equations of motion are

$$\dot{\beta}_s = \sqrt{\mathcal{P}} - (1 + \mathrm{i}\tilde{\Delta} + |\beta_i|^2)\beta_s + \varepsilon\beta_i^* \tag{5.6a}$$

$$\dot{\beta}_i = \sqrt{\mathcal{P}} - (1 - \mathrm{i}\tilde{\Delta} + |\beta_s|^2)\beta_i + \varepsilon\beta_s^* \tag{5.6b}$$

These equations possess the symmetry $\{\beta_s \to \beta_i^*, \beta_i \to \beta_s^*\}$, which suggests looking for stationary solutions of the form

$$\bar{\beta}_s = \bar{\beta}_i^* = \sqrt{I}\exp(\mathrm{i}\phi), \quad \text{with } I \in \mathbb{R} > 0 \text{ and } \phi \in \mathbb{R}. \tag{5.7}$$

Substituting this in (5.6) we find

$$\sqrt{\mathcal{P}} = (I + 1 - \varepsilon + \mathrm{i}\tilde{\Delta})\sqrt{I}\exp(\mathrm{i}\phi), \tag{5.8}$$

which can be written as

$$\sqrt{\mathcal{P}} = |I + 1 - \varepsilon + \mathrm{i}\Delta|\sqrt{I}\exp\left(\mathrm{i}\arg\{I + 1 - \varepsilon + \mathrm{i}\tilde{\Delta}\} + \mathrm{i}\phi\right). \tag{5.9}$$

Since $\sqrt{\mathcal{P}}$ is real and positive, $\phi = -\arg\{I + 1 - \varepsilon + \mathrm{i}\tilde{\Delta}\}$. On the other hand, the intensity I of this symmetric solution satisfies the third order polynomial

$$\mathcal{P} = [(I + 1 - \varepsilon)^2 + \tilde{\Delta}^2]I \tag{5.10}$$

Depending on the parameters this polynomial can have one or three positive solutions. By finding the turning points $I_\pm$ satisfying $\partial\mathcal{P}/\partial I|_{I=I_\pm} = 0$, we can find the region of parameters where these alternatives occur. We get

$$I_\pm = \frac{2}{3}(\varepsilon - 1) \pm \frac{1}{3}\sqrt{(\varepsilon - 1)^2 - 3\tilde{\Delta}^2}, \tag{5.11}$$

which shows that $I_\pm$ are real and positive only as long as $\varepsilon \geq 1 + \sqrt{3}\tilde{\Delta}$, which defines the domain where three stationary solutions coexist in the bifurcation diagram, see Figs. 5.1 and 5.2. For smaller values of the pump ε, the stationary solution is unique.

Not all stationary solutions are stable (see Fig. 5.1), and to see which of these stationary solutions are stable, we now turn our attention to the linear stability analysis discussed in the previous section. It is convenient to move to a polarization basis which exploits the symmetry of the stationary solution (5.7), which will provide a linear stability matrix that is simpler to work with. The new modes are defined by

$$\beta_b = \frac{e^{-\mathrm{i}\phi}\beta_s + e^{\mathrm{i}\phi}\beta_i}{\sqrt{2}}, \qquad \beta_d = \mathrm{i}\frac{e^{-\mathrm{i}\phi}\beta_s - e^{\mathrm{i}\phi}\beta_i}{\sqrt{2}}, \tag{5.12}$$

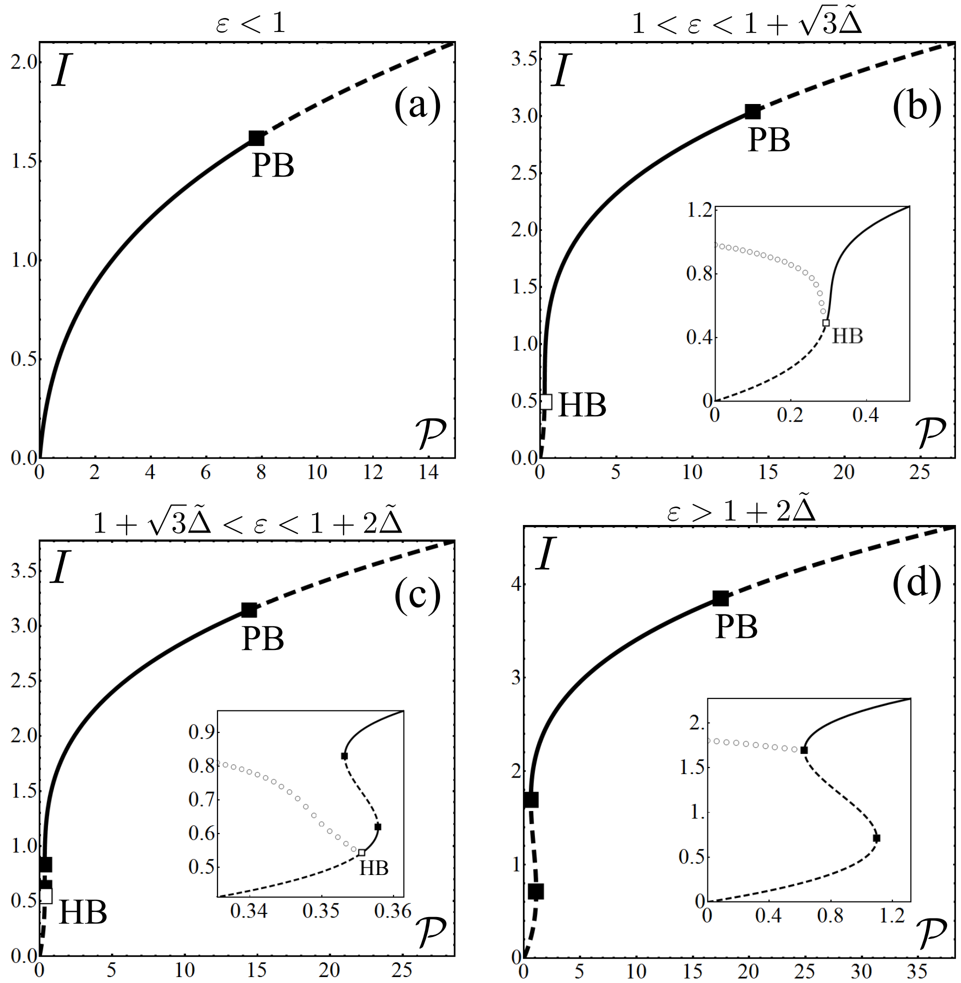

FIGURE 5.1. Bifurcation diagram of the system. The value $\tilde{\Delta} = 0.6$ is chosen for all the figures (the same behavior is found for any other choice), while we set ε to 0.5 in top-left, 1.98 in top-right, 2.09 in bottom-left, and 2.8 in bottom-right. The black lines correspond to the intensity I of the stationary symmetric solution (5.7), their solid or dashed character meaning that this solution is stable or unstable, respectively. As explained in the text, for $\varepsilon > 1$ it is possible to find periodic solutions connecting the $\mathcal{P} = 0$ axis with the stationary branch (at the Hopf bifurcation, marked as HB, for $\varepsilon < 1 + 2\tilde{\Delta}$ or the upper turning point otherwise). We have checked numerically that these periodic orbits exist, and moreover they are 'symmetric', that is, $\beta_s(\tau) = \beta_i^*(\tau)$. The gray circles correspond in this case to the mean value of $|\beta_s(\tau)|^2$ (half the sum between its maximum and its minimum of oscillation). On the other hand, for relatively large injections there is always a Pitchfork bifurcation (PB in the plot) in which the symmetric stationary solution becomes unstable in favor of another stationary but asymmetric solution. The insets show simply a zoom of the main figures at low injections. Figure taken from [9].

where the labels refer to *bright* and *dark*, since in this basis the stationary solution (5.7) reads ($\bar{\beta}_b = \sqrt{2}I, \bar{\beta}_d = 0$). In this new variables, the classical equations (5.6) read

$$\dot{\beta}_b = \sqrt{2\mathcal{P}}\cos\phi - \beta_b - \tilde{\Delta}\beta_d + \left(\varepsilon - \frac{\beta_b^2}{2} - \frac{\beta_d^2}{2}\right)\beta_b^*, \tag{5.13}$$

$$\dot{\beta}_d = \sqrt{2\mathcal{P}}\cos\phi - \beta_d + \tilde{\Delta}\beta_b + \left(\varepsilon - \frac{\beta_b^2}{2} - \frac{\beta_d^2}{2}\right)\beta_d^*. \tag{5.14}$$

Using the order $(\beta_b, \beta_b^*, \beta_d, \beta_d^*)$, the linear stability matrix associated to the symmetric stationary solution is

$$\mathcal{L} = \begin{bmatrix} -1-2I & \varepsilon - I & -\tilde{\Delta} & 0 \\ \varepsilon - I & -1-2I & 0 & -\tilde{\Delta} \\ \tilde{\Delta} & 0 & -1 & \varepsilon - I \\ 0 & \tilde{\Delta} & \varepsilon - I & -1 \end{bmatrix} \tag{5.15}$$

It is interesting to note that the bright and dark subspaces are only coupled through the detuning $\tilde{\Delta}$. As discussed in the previous section, the eigenvalues of the linear stability matrix determine the stability of the symmetric stationary solution. In the case of this matrix, the eigenvalues read

$$\lambda_\pm^I = -(1+\varepsilon) \pm \sqrt{I^2 - \tilde{\Delta}^2}, \tag{5.16a}$$

$$\lambda_\pm^{II} = \varepsilon - 1 - 2I \pm \sqrt{I^2 - \tilde{\Delta}^2} \tag{5.16b}$$

In Fig. 5.2 we summarize the behavior of these eigenvalues, by showing the regions in the parameter space where their real and imaginary parts have different character. Note that we use I as a parameter instead of $\mathcal{P}$, because the latter is uniquely defined by the former from (5.10), but not the other way around. The lines separating stable regions ($\mathrm{Re}\{\lambda_\pm^I, \lambda_\pm^{II}\} < 0$) from unstable regions are the bifurcations we are seeking for, where the real part of at least one eigenvalue vanishes.

Looking at the first pair of eigenvalues, we find a pitchfork bifurcation $\lambda_+^I = 0$ when the intensity takes the value

$$I_{PB} = \sqrt{(1+\varepsilon)^2 + \tilde{\Delta}^2}, \tag{5.17}$$

which exists for any value of ε and $\tilde{\Delta}$. Note that this value is larger than I_+, and therefore this bifurcation is located in the upper branch of the S-shaped curve of Fig. 5.1. For intensities larger than this value, the symmetric solution becomes unstable, and a new stationary solution with $\bar{\beta}_d \neq 0$ or, equivalently, $\bar{\beta}_s \neq \bar{\beta}_i^*$, is born. No other bifurcations are found from this eigenvalues.

We turn now our attention to the second pair of eigenvalues. First, it is easy to show that $\lambda_\pm^{II} = 0$ for $I = I_\pm$, which are the turning points we found above. Therefore, these

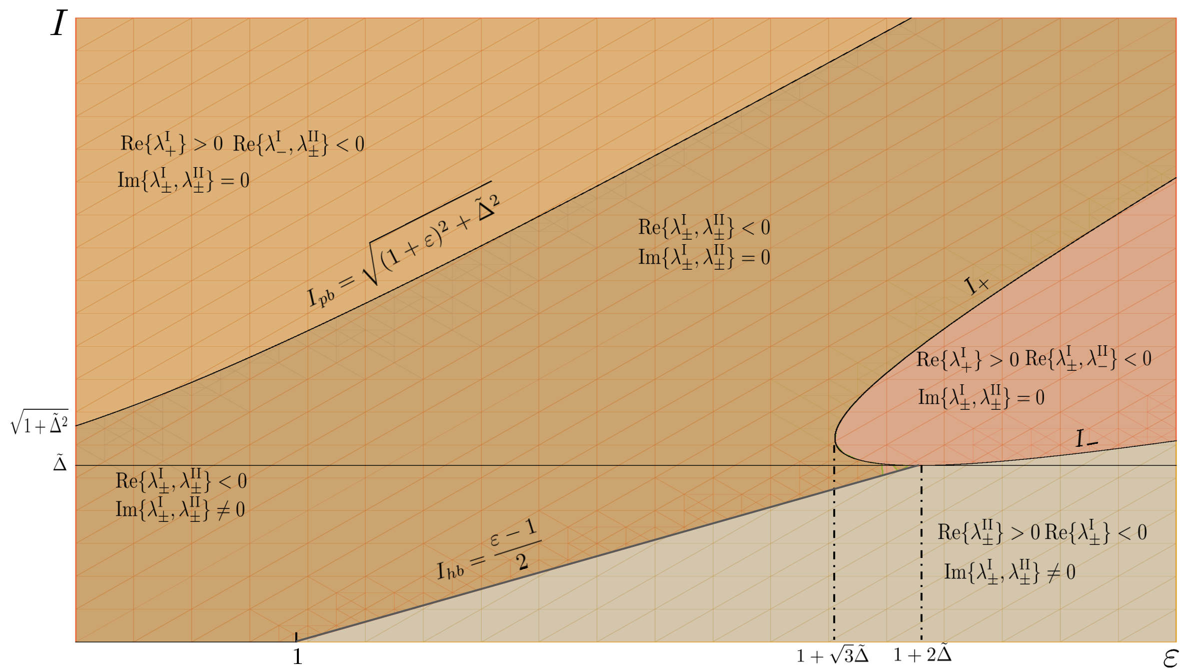

FIGURE 5.2. [Make black line dashed.]Phase diagram of the system in the classical limit, based on the stability properties of the symmetric solution (5.7). The value $\tilde{\Delta}^2 = 2$ is chosen for this diagram, but the same behaviour is found for any other choice. Note that, for convenience, we use I (intracavity intensity of the symmetric solution) as a parameter instead of the actual external power $\mathcal{P}$, but the latter is uniquely defined from the former by (5.10). The $I_\pm$ curves correspond to the turning points (5.11) of the S-shape curve in Fig. 5.1. In the region enclosed by these curves the solution is unstable, which corresponds to the middle branch of the S-shaped curve as can also be seen in Fig. 5.1. Above the line $I = \tilde{\Delta}$ the eigenvalues are real, while below they have a non-zero imaginary part. The line I_{HB} corresponds to Hopf bifurcation, which connects limit cycles to the stable stationary solutions. The curve I_{PB} corresponds to a pitchfork bifurcation which connects the symmetric stationary solution to an asymmetric one.

turning points are bifurcations, and in particular they are easily shown to signal that the middle branch of the S-shaped curve in Fig. 5.1 is unstable. In other words, whenever there is coexistence of three solutions, the one with intensity in the middle is unstable.

But $\lambda_\pm^{II}$ is affected by one more instability, a Hopf bifurcation in particular, when the intensity reaches the value

$$I_{HB} = \frac{\varepsilon - 1}{2} \tag{5.18}$$

which clearly requires $\varepsilon > 1$. At this point, the eigenvalue becomes purely imaginary $\lambda_\pm^{II} = \pm i\omega_{HB}$ with a Hopf frequency $\omega_{HB} = \sqrt{\tilde{\Delta}^2 - (\varepsilon - 1)^2/4}$ that requires $\varepsilon < 1 + 2\tilde{\Delta}$.

Hence, this bifurcation is present only when $1 < \varepsilon < 1 + 2\tilde{\Delta}$. Specifically, for $\varepsilon = 1$ the bifurcation is located at $I = 0$, and it climbs up the lower branch of the S-shaped curve as we increase ε, until it disappears for $\varepsilon = 1 + 2\tilde{\Delta}$ when it coalesces with the lower turning point. When it exists, the portion of the lower branch with $I < I_{HB}$ is unstable, and because this is a Hopf bifurcation we expect the new asymptotically stable solution to correspond to limit cycles, what we checked numerically as can be appreciated in Fig. 5.1. When the Hopf bifurcation ceases to exist for $\varepsilon > 1 + 2\tilde{\Delta}$, the limit cycles still exist but then connect with the upper turning point (see Fig. 5.1 bottom-right), and the whole lower branch becomes unstable.

This analysis proves that there exist region in the parameter space where the frequencies of the signal and idler modes are locked to the degenerate one (the regions where the stationary solution is stable). The quantum properties at such regions were studied in [9]. For the purposes of this thesis, we are interested in extending this quantum analysis to the regions of limit-cycle motion, which we will do along the next chapters.

CHAPTER 6

QUANTUM FLUCTUATIONS AROUND LIMIT CYCLES

We will now concentrate on the description of the quantum state in the region of parameters where the classical asymptotic state corresponds to a limit cycle. We will proceed by considering small quantum fluctuations around such classical state, the so-called *linearized theory*. However, as compared with simple stationary classical solutions, two difficulties appear when trying to apply the method to limit cycles. First, the linear stability matrix is not constant, but time-periodic. At this point we might wonder whether we can still find some sort of periodic eigenvectors, whose corresponding eigenvalues inform us about the stability of the limit cycle and allow us to solve the linearized problem efficiently. This is precisely what Floquet theory offers, which we will apply in our theory. The second difficulty stems from the fact that the limit cycle is not a single point in phase space, but a continuous set of points which form a closed trajectory. Hence, there must always be a direction along which fluctuations are not restricted or dampened (corresponding to a zero eigenvalue), so that we cannot assume that they are small, rendering the linearization technique inconsistent and flawed. We will go around this problem by explicitly identifying the system's variable that gets the larger part of the unbounded fluctuations, linearizing then only with respect to the rest of the variables. This method, which was first introduced in [10] will allow us to provide an approximate expression for the quantum steady state of the system as a mixture of Gaussian states.

6.1 Initial considerations and outline of our goal

Let us start by introducing some general considerations about the state of the system in the classical limit, and how the method that we will introduce later goes beyond it. We stick here to a single-mode description for simplicity, but everything is trivially generalized to any number of modes.

Consider a choice of parameters for which the system reaches limit cycle motion, which we denote by $\bar{\alpha}(t)$ with periodicity T, such that this corresponds to a closed curve in phase space. In other words, if the system is found at some point $\bar{\alpha}(0)$ at $t = 0$, we know it should be back at that point at times $t = nT$ with $n \in \mathbb{N}$. This is only true in the absence of noise, and now let's try to understand the effect that noise will have in the system.

First, note that trying to kick the system out of the phase-space trajectory defined by the cycle is met with a quick reaction from the system pulling itself back to the trajectory (that's what it means for the cycle to be stable), similarly to what happens with a stable stationary solution. In contrast, performing arbitrary translations along the cycle's trajectory in phase space is allowed without any resistance from the system, since $\bar{\alpha}(t + \theta)$ is a limit-cycle solution of the classical equations of motion for any choice of the constant θ. If there was no noise, starting from some initial condition the system would eventually fall into the limit cycle, choosing the value of θ according to whatever random initial conditions were set. We say that the system undergoes spontaneous symmetry breaking, since the solution $\bar{\alpha}(t + \theta)$ is not invariant under time translations, even though this is a symmetry of the equations of motion. Now, when noise is included, its projection orthogonal to the trajectory is dampen quickly, but the one along the cycle's trajectory is allowed to push around the system's motion in any random way it wants. This can be described mathematically as a translation θ that turns more and more unpredictable as time goes by, so that the state of the system is no longer a well defined coherent state $|\bar{\alpha}(t)\rangle$ cycling along the trajectory, but a mixture of all the possible ones which start cycling at the different points of the cycle, a situation represented by the mixed state:

$$\lim_{t\to\infty} \hat{\rho}(t) = \int_0^T \frac{dt}{T} |\bar{\alpha}(t)\rangle\langle\bar{\alpha}(t)|. \tag{6.1}$$

This is just a way of saying that once noise drives the system into its asymptotic steady state, we have no way of predicting where the system will be along the cycle's trajectory. Therefore, this is the quantum steady state expected when working in the classical limit, which as explained at the end of chapter 4, corresponds to small g (see also [10]).

Our goal in the next sections is finding a better approximation for this stationary quantum state, specifically one which incorporates quantum fluctuations beyond those of a coherent state. In particular, instead of a mixture of coherent states along the cycle's trajectory, we will find some more general type of Gaussian states $\hat{\rho}_G(t)$ at each point of the trajectory, which will approximate better the stationary quantum state of the system for finite $g \neq 0$ as

$$\lim_{t\to\infty} \hat{\rho}(t) = \int_0^T \frac{dt}{T} \hat{\rho}_G(t). \tag{6.2}$$

This was already done in [10] for the example of the quantum Van der Pol oscillator model, but here we apply it to a more physically and experimentally relevant scenario, the actively phase-locked type-II OPO. Moreover, while the Gaussian states appearing in the Van der Pol model were rather trivial, the ones that will appear in the OPO showcase many interesting features that we will study in detail in the next chapter. First, they will be shown to predict entanglement between signal and idler. Second, they will be shown to incur into an apparent violation of the uncertainty principle. Of course, the overall mixed state (6.2) will still be physical, and we will discuss the effect that this might have on the entanglement.

6.2 Linearization around limit cycles

Let us now proceed to apply the linearization technique to the Langevin equations (4.46) of our OPO. In the following we denote by $\bar{\beta}_j(\tau)$ the limit cycle solutions.

Naively, one would simply proceed by considering small fluctuations around the classical limit cycle, e.g., $\beta_j(\tau) = \bar{\beta}_j(\tau) + \delta\beta_j(\tau)$, linearizing then with respect to the fluctuations. However, as explained in the previous section, because translations along the cycle's trajectory are not dampen, the fluctuations cannot be considered small along such trajectory of phase space. Mathematically, this would appear through a zero (Floquet) eigenvalue of the linear stability matrix, that would predict a variance of the fluctuations that increases with time, hence leaving the "small- fluctuation" regime and generating nonsensical predictions.

As showed in [10], this can be circumvented by introducing a variable that explicitly tracks the larger part of the fluctuations, and is related to the symmetry of the equations that is spontaneously broken by the classical solution [24, 25]: time translations in the case at hand. We then expand the stochastic amplitudes as

$$\beta_j\big(\tau + \theta(t)\big) = \bar{\beta}_j\big(\tau + \theta(t)\big) + b_j\big(\tau + \theta(t)\big), \tag{6.3a}$$

$$\beta_j^+\big(\tau + \theta(t)\big) = \bar{\beta}_j^*\big(\tau + \theta(t)\big) + b_j^+\big(\tau + \theta(t)\big), \tag{6.3b}$$

As we will show later, $\theta(\tau)$ will carry the part of the fluctuations that increases with time, so that the fluctuations (b, b^+) as well as the noises (η, η^+) can be considered small in a consistent fashion, following the standard recipe of linearization around stationary solutions (more rigorously, we are considering an expansion to linear order in $\sqrt{g}$). Moreover, since the rate of change of θ is expected to be proportional to quantum noise, $\dot{\theta}$ can also be considered small. Introducing (6.3) into the stochastic Langevin equations (4.46) and ignoring terms beyond first order in the small variables, it is possible [10] to obtain the linear system

$$\dot{\boldsymbol{b}}(\tau) + \boldsymbol{p}_0(\tau)\dot{\theta}(\tau) = \mathcal{L}(\tau)\boldsymbol{b}(\tau) + \sqrt{g}\,\boldsymbol{\xi}(\tau), \tag{6.4}$$

where we have defined the vectors $\boldsymbol{b} = (b_s, b_s^+, b_i, b_i^+)^{\mathrm{T}}$, $\boldsymbol{p}_0(\tau) = (\partial_\tau \bar{\beta}_s, \partial_\tau \bar{\beta}_s^*, \partial_\tau \bar{\beta}_i, \partial_\tau \bar{\beta}_i^*)^{\mathrm{T}}$, and $\boldsymbol{\xi}(\tau) = \sqrt{\varepsilon - |\bar{\beta}_s(\tau)|^2}(\eta(\tau), \eta^+(\tau), \eta^*(\tau), \eta^{+*}(\tau))^{\mathrm{T}}$, while $\mathcal{L}(\tau)$ is the linear stability matrix evaluated at the limit cycle,

$$\mathcal{L}(\tau) = -\begin{pmatrix} 1 + \mathrm{i}\tilde{\Delta} + |\bar{\beta}_i(\tau)|^2 & 0 & \bar{\beta}_s(\tau)\bar{\beta}_i^*(\tau) & \bar{\beta}_s(\tau)\bar{\beta}_i(\tau) - \varepsilon \\ 0 & 1 - \mathrm{i}\tilde{\Delta} + |\bar{\beta}_i(\tau)|^2 & \bar{\beta}_s^*(\tau)\bar{\beta}_i^*(\tau) - \varepsilon & \bar{\beta}_s^*(\tau)\bar{\beta}_i(\tau) \\ \bar{\beta}_i(\tau)\bar{\beta}_s^*(\tau) & \bar{\beta}_i(\tau)\bar{\beta}_s(\tau) - \varepsilon & 1 + \mathrm{i}\tilde{\Delta} + |\bar{\beta}_s(\tau)|^2 & 0 \\ \bar{\beta}_s^*(\tau)\bar{\beta}_i^*(\tau) - \varepsilon & \bar{\beta}_s(\tau)\bar{\beta}_i^*(\tau) & 0 & 1 - \mathrm{i}\tilde{\Delta} + |\bar{\beta}_s(\tau)|^2 \end{pmatrix} \tag{6.5}$$

which inherits the cycle's periodicity T, that is, $\mathcal{L}(\tau) = \mathcal{L}(\tau + T)$. Let us remark here that the limit cycle $\{\bar{\beta}_s(\tau), \bar{\beta}_i(\tau)\}$ can only be found numerically in general. Also, after an exhaustive analysis, we can conclude that the limit cycles follow the symmetry of the "symmetric" solution, that is, $\bar{\beta}_s(\tau) = \bar{\beta}_i^*(\tau)$.

6.3 Floquet method

The main difference between (6.4) and the standard linearized equations that appear around stationary solutions is the time-dependence of the linear stability matrix. This makes it difficult to solve the equations to evaluate any observable quantities we might need, but now we show that Floquet theory simplifies this task immensely.

Floquet theory start by defining the fundamental matrix $\mathcal{R}(\tau)$, which satisfies the initial value problem $\dot{\mathcal{R}}(\tau) = \mathcal{L}(\tau)\mathcal{R}(\tau)$ with $\mathcal{R}(0) = \mathcal{I}$, where $\mathcal{I}$ is the identity matrix. From it, we further define the constant matrix $\mathcal{M}$ through $e^{\mathcal{M}T} = \mathcal{R}(T)$, and the T-periodic matrix $\mathcal{P}(\tau) = \mathcal{R}(\tau)e^{-\mathcal{M}\tau}$. Given the eigensystem $\{\boldsymbol{v}_j, \boldsymbol{w}_j; \mu_j\}_{j=0..3}$ of $\mathcal{M}$, composed of right and left orthogonal ($\boldsymbol{w}_j^\dagger \boldsymbol{v}_i = \delta_{ij}$) eigenvectors satisfying $\mathcal{M}\boldsymbol{v}_j = \mu_j \boldsymbol{v}_j$ and $\boldsymbol{w}_j^\dagger \mathcal{M} = \mu_j \boldsymbol{w}_j^\dagger$, we introduce the Floquet eigenvectors $\boldsymbol{p}_j(\tau) = \mathcal{P}(\tau)\boldsymbol{v}_j$ and $\boldsymbol{q}_j^\dagger(\tau) = \boldsymbol{w}_j^\dagger \mathcal{P}^{-1}(\tau)$. The Floquet vectors will be used to define fluctuations along relevant directions along the limit cycle. By using the above mentioned relations we can show that the Floquet vectors satisfy the initial value problems

$$\dot{\boldsymbol{p}}_j(\tau) = [\mathcal{L}(\tau) - \mu_j]\boldsymbol{p}_j(\tau), \quad \boldsymbol{p}_j(0) = \boldsymbol{v}_j, \tag{6.6}$$

$$\dot{\boldsymbol{q}}_j^\dagger(\tau) = \boldsymbol{q}_j^\dagger(\tau)[\mu_j - \mathcal{L}(\tau)], \quad \boldsymbol{q}_j^\dagger(0) = \boldsymbol{w}_j^\dagger \tag{6.7}$$

and the orthogonality condition $\boldsymbol{q}_j^\dagger(\tau)\boldsymbol{p}_l(\tau) = \delta_{jl} \forall \tau$.

We mentioned above that there always exists a direction in phase space along which the evolution of perturbations is not dampened. Now we can show explicitly that this is the case, since there is a Floquet eigenvector with zero eigenvalue. Specifically, it is easy to show that the vector $\mathrm{p}_0(\tau)$ defined in (6.4) satisfies the equation $\dot{\boldsymbol{p}}_0(\tau) = \mathcal{L}(\tau)\boldsymbol{p}_0(\tau)$, which has the form (6.6) with $\mu_0 = 0$, and hence it is a right Floquet eigenvector with zero eigenvalue. The detailed proof can be found in [10].

Apart from $\mathbf{p}_0$, the rest of eigenvectors are found numerically by applying the Floquet procedure explained above. Note, however, that based on [10] we have the analytic form of another Floquet eigenvector, a left one specifically, but we don't discuss it here because it gives us no special benefit, since all the rest must be found numerically anyways.

6.4 A convenient change of basis

Let's now go back to our linear system of interest given in (6.4). The linear stability matrix of our system is 4×4, which is not easy to work with. However, next we make a variable change that will lead to two independent linear systems with 2×2 linear stability matrices. Specifically, let us define the new variables

$$\beta_\pm = \beta_s \pm \beta_i^+, \quad \beta_\pm^+ = \beta_s^+ \pm \beta_i \tag{6.8}$$

Note that this new variables do not correspond to any specific physical modes, since the transformation is not canonical, as the corresponding operators $\{\hat{a}_s \pm \hat{a}_i^\dagger, \hat{a}_s^\dagger \pm \hat{a}_i\}$ do not satisfy canonical commutation relations. In any case, since the variable change is linear, it is always possible to come back to the original signal-idler variables, as we will do later.

In terms of this new variables, the Langevin equations (4.46) can be written as

$$\begin{aligned}
\dot{\beta}_+ &= 2\sqrt{\mathcal{P}} + (\varepsilon - 1 - \mathrm{i}\Delta)\beta_+ - \frac{\beta_+^+}{4}(\beta_+^2 - \beta_-^2) && \text{(6.9a)}\\
&\quad + \sqrt{\frac{g}{4}\big(\varepsilon - (\beta_+ + \beta_-)(\beta_+^+ + \beta_-^+)\big)}\,\eta(\tau) + \sqrt{\frac{g}{4}\big(\varepsilon - (\beta_+ - \beta_-)(\beta_+^+ - \beta_-^+)\big)}\,\eta^{+*}(\tau),\\
\dot{\beta}_+^+ &= 2\sqrt{\mathcal{P}} + (\varepsilon - 1 + \mathrm{i}\Delta)\beta_+^+ - \frac{\beta_+}{4}(\beta_+^{+2} - \beta_-^{+2}) && \text{(6.9b)}\\
&\quad + \sqrt{\frac{g}{4}\big(\varepsilon - (\beta_+^+ + \beta_-^+)(\beta_+ + \beta_-)\big)}\,\eta^+(\tau) + \sqrt{\frac{g}{4}\big(\varepsilon - (\beta_+ - \beta_-)(\beta_+^+ - \beta_-^+)\big)}\,\eta^*(\tau),\\
\dot{\beta}_- &= (-\varepsilon - 1 - \mathrm{i}\Delta)\beta_- - \frac{\beta_-^+}{4}(\beta_-^2 - \beta_+^2) && \text{(6.9c)}\\
&\quad + \sqrt{\frac{g}{4}\big(\varepsilon - (\beta_+ + \beta_-)(\beta_+^+ + \beta_-^+)/4\big)}\,\eta(\tau) - \sqrt{\frac{g}{4}\big(\varepsilon - (\beta_+ - \beta_-)(\beta_+^+ - \beta_-^+)\big)}\,\eta^{+*}(\tau),\\
\dot{\beta}_-^+ &= (-\varepsilon - 1 + \mathrm{i}\Delta)\beta_-^+ - \frac{\beta_-}{4}(\beta_-^{+2} - \beta_+^{+2}) && \text{(6.9d)}\\
&\quad + \sqrt{\frac{g}{4}\big(\varepsilon - (\beta_+ + \beta_-)(\beta_+^+ + \beta_-^+)\big)}\,\eta^+(\tau) - \sqrt{\frac{g}{4}\big(\varepsilon - (\beta_+ - \beta_-)(\beta_+^+ - \beta_-^+)\big)}\,\eta^*(\tau).
\end{aligned}$$

Because of the property $\bar{\beta}_s(\tau) = \bar{\beta}_i^*(\tau)$ of the classical limit-cycle solutions, in the new variable the stationary solutions are $\bar{\beta}_-(\tau) = 0$ and $\bar{\beta}_+(\tau) = 2\bar{\beta}_s(\tau)$. In addition, unlike in the bright/dark basis introduced when analyzing the stability of the classical steady-state solution, there is no linear coupling between β_+ and β_-. Therefore, linearizing these equations around the classical limit cycle decouples the "plus" and "minus" subspaces.

Specifically, using the expansion (6.3) with $j = \pm$, we obtain the decoupled linear systems

$$\dot{\boldsymbol{b}}_+(\tau) + \boldsymbol{p}_0(\tau)\dot{\theta}(\tau) = \mathcal{L}_+(\tau)\boldsymbol{b}_+(\tau) + \sqrt{g}\,\boldsymbol{\xi}_+(\tau) \tag{6.10a}$$

$$\dot{\boldsymbol{b}}_-(\tau) = \mathcal{L}_-(\tau)\boldsymbol{b}_-(\tau) + \sqrt{g}\,\boldsymbol{\xi}_-(\tau) \tag{6.10b}$$

where (note that we keep the notation $\boldsymbol{p}_0$ for the corresponding Floquet eigenvector, even if we write it in the new space, because we won't need to differentiate these representations)

$$\boldsymbol{b}_\pm(\tau) = \left(b_\pm(\tau), b_\pm^+(\tau)\right)^{\mathrm{T}}, \tag{6.11a}$$

$$\boldsymbol{p}_0(\tau) = \left(\partial_t\bar{\beta}_+(\tau), \partial_t\bar{\beta}_+^*(\tau)\right)^{\mathrm{T}}, \tag{6.11b}$$

$$\boldsymbol{\xi}_\pm(\tau) = \sqrt{\varepsilon - |\bar{\beta}_+|^2/4}\left(\eta(\tau) \pm \eta^{+*}(\tau), \eta^+(\tau) \pm \eta^*(\tau)\right)^{\mathrm{T}}, \tag{6.11c}$$

and we have defined the linear stability matrices

$$\mathcal{L}_+(\tau) = \begin{pmatrix} \varepsilon - 1 - \mathrm{i}\tilde{\Delta} - |\bar{\beta}_+(\tau)|^2/2 & -\bar{\beta}_+^2/4 \\ -\bar{\beta}_+^{*2}/4 & \varepsilon - 1 + \mathrm{i}\tilde{\Delta} - |\bar{\beta}_+(\tau)|^2/2 \end{pmatrix} \tag{6.12a}$$

$$\mathcal{L}_-(\tau) = \begin{pmatrix} -\varepsilon - 1 - \mathrm{i}\tilde{\Delta} & \bar{\beta}_+^2/4 \\ \bar{\beta}_+^{*2}/4 & -\varepsilon - 1 + \mathrm{i}\tilde{\Delta} \end{pmatrix}, \tag{6.12b}$$

Hence, we see that we can treat the "plus" and "minus" subspaces separately, simplifying our four-dimensional problem into two two-dimensional ones.

6.5 Diffusion of θ and covariance matrix of the system

We would now like to analyze the fluctuations along the Floquet vectors of the "plus" and "minus" subspaces. We will denote the Floquet eigensystem corresponding to $\mathcal{L}_+(\tau)$ by $\{\boldsymbol{p}_j, \boldsymbol{q}_j; \mu_j\}_{j=0,1}$, which we find numerically by employing the Floquet method explained above (specifically, we find the fundamental matrix by solving numerically its initial value problem from $\tau = 0$ to $\tau = T$, and then proceed with the rest of definitions). Similarly, we denote by $\{\boldsymbol{p}_j, \boldsymbol{q}_j; \mu_j\}_{j=2,3}$ the Floquet eigensystem of $\mathcal{L}_-(\tau)$. Applying the left eigenvectors onto (6.10a) and (6.10b), and using the general properties of the Floquet eigenvectors, we easily find the equations

$$\frac{d}{d\tau}(c_0 + \theta) = \sqrt{g}\,\boldsymbol{q}_0^\dagger(\tau)\boldsymbol{\xi}_+(\tau), \tag{6.13a}$$

$$\dot{c}_j = \mu_j c_j + \sqrt{g}\,\boldsymbol{q}_j^\dagger(\tau)\boldsymbol{\xi}_+(\tau) \qquad j = 1, 2, 3, \tag{6.13b}$$

where we have defined the projections $c_{0,1}(\tau) = \boldsymbol{q}_{0,1}^\dagger(\tau)\boldsymbol{b}_+(\tau)$ and $c_{2,3}(\tau) = \boldsymbol{q}_{2,3}^\dagger(\tau)\boldsymbol{b}_-(\tau)$, and used the fact that $\mu_0 = 0$.

Since by using the expansion (6.3) we introduced a new variable $\theta(\tau)$ into the equations, there is an excess (redundancy) of one variable which we are now able to identify as $c_0(\tau)$, and consistently set to zero. In other words, without introducing $\theta(\tau)$, we would have simply obtained a projection $c_0(\tau)$ of the fluctuations whose variance becomes arbitrarily large with time, hence destroying the linear approximation. However, we understand now that $c_0(\tau)$ is simply another way of denoting the fluctuations along the cycle's trajectory, which we have introduced from the start explicitly as $\theta(\tau)$. Setting then $c_0(\tau) = 0$ and solving (6.13a), we find variance of $\theta(\tau) - \theta(0)$ to be

$$V_\theta(\tau) = \left\langle \left(\theta(\tau) - \theta(0)\right)^2 \right\rangle = g \int_0^\tau d\tau' \boldsymbol{q}_0^\dagger(\tau') D(\tau') \boldsymbol{q}_0^*(\tau') \tag{6.14}$$

where the matrix $D(\tau)$ encode the noise correlations as

$$\langle \boldsymbol{\xi}_\pm(\tau) \boldsymbol{\xi}_\pm^{\mathrm{T}}(\tau') \rangle = \underbrace{\begin{pmatrix} 0 & 2\varepsilon - |\bar{\beta}_+|^2/2 \\ 2\varepsilon - |\bar{\beta}_+|^2/2 & 0 \end{pmatrix}}_{D(\tau)} \delta(\tau - \tau'). \tag{6.15}$$

The kernel of the integral is periodic, and the value of the integral is different than zero over one period. Hence, the variance of θ increases linearly with the number of periods that motion has undergone, as appreciated in the Fig. (6.1). In other words, as time passes, θ becomes more and more undetermined. This fits well with our expectations about what noise (whether quantum or classical) should do, as we explained in the first section of the chapter. Indeed, we see that the variance vanishes if we go to the strictly deterministic limit $g = 0$, just as expected.

In contrast, we expect the fluctuations in any other direction of phase space to be finite and small. Next we show this by solving the rest of the equations (6.13b), and showing that the asymptotic covariance matrix of the projections, denoted by $C(\tau)$, with elements

$$C_{jl}(\tau) = \lim_{\tau\to\infty} \langle c_j(\tau) c_l(\tau) \rangle, \tag{6.16}$$

doesn't diverge with time. Note that here we are using the notation $\lim_{\tau\to\infty}$ loosely as "asymptotic value for long times", such that $\lim_{\tau\to\infty} f(\tau)$ can depended on τ if $f(\tau)$ converges to some time periodic solution, as is our case in general. Assuming that all the Floquet eigenvalues (other than μ_0) have a negative real part (as corresponds to stable limit cycles), the asymptotic solution of (6.13b) reads

$$c_j(\tau) = \sqrt{g} \lim_{\tau\to\infty} \int_0^\tau d\tau' \boldsymbol{q}_j^\dagger(\tau') \boldsymbol{\xi}_+(\tau'), \tag{6.17}$$

leading to an asymptotic zero average, $\lim_{\tau\to\infty} \langle c_j(\tau) \rangle = 0$, and a covariance matrix

$$C_{jl}(\tau) = g \lim_{\tau\to\infty} \int_0^\tau d\tau' e^{(\mu_j+\mu_l)(\tau-\tau')} \boldsymbol{q}_j^\dagger(\tau') D(\tau') \boldsymbol{q}_l^*(\tau'). \tag{6.18}$$

Note three things about this expression. First, the Kernel of the integral is not time-periodic, but decays with time, and hence the integral cannot diverge as time increases.

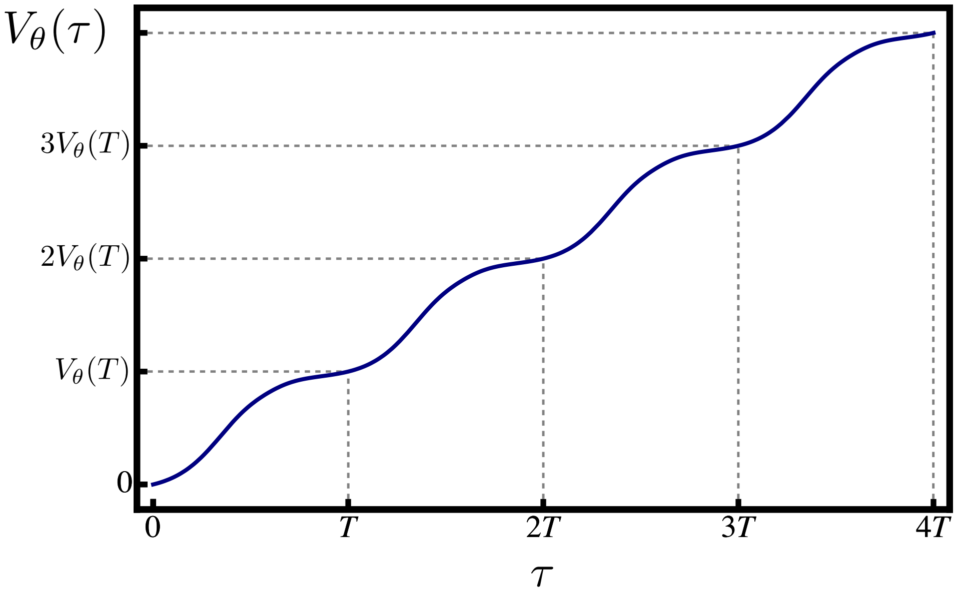

FIGURE 6.1. Time evolution of the variance of θ (6.14). We have chosen a limit cycle with parameters $\tilde{\Delta} = 0.1$, $\varepsilon = 1 + \sqrt{3}\tilde{\Delta}/2$, and $I = I_{HB}/2$ corresponding to $\mathcal{P} = 19\sqrt{3}\tilde{\Delta}^3/64$, which is in the middle of the upper-right panel of fig. 5.1. For this cycle, we obtain $V_\theta(T) \approx 6g \times 10^6$, where g is the parameter that controls the strength of the quantum noise.

On the other hand, it is easy to prove that $C(\tau)$ is T-periodic. Finally, we see that it is proportional to g, which indeed controls the strength of the quantum noise, meaning that quantum fluctuations in the normalized variables will remain small as long as g is small, which is consistent with the initial linearized approximation.

Once we have figured out the first and second moments of the projections, which we collect in the vector $\boldsymbol{c} = (0, c_1, c_2, c_3)^{\mathrm{T}}$ for later convenience, we can finding any other quantity of interest. For example, let us start by find the covariance matrix of the $\beta_\pm$ and $\beta_\pm^+$ variables, whose fluctuations we collect in the vector $\boldsymbol{b}_\pm = (b_+, b_+^+, b_-, b_-^+)^{\mathrm{T}}$. Defining the 4×4 matrix

$$Q(\tau) = \begin{pmatrix} \boldsymbol{q}_0^\dagger(\tau) & \boldsymbol{0} \\ \boldsymbol{q}_1^\dagger(\tau) & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{q}_2^\dagger(\tau) \\ \boldsymbol{0} & \boldsymbol{q}_3^\dagger(\tau) \end{pmatrix}, \tag{6.19}$$

where $\boldsymbol{0} = (0, 0)$, the definition of the projections can be recasted in the matrix form $\boldsymbol{c}(\tau) = Q(\tau)\boldsymbol{b}_\pm(\tau)$, which we invert as

$$\boldsymbol{b}_\pm(\tau) = Q^{-1}(\tau)\boldsymbol{c}(\tau). \tag{6.20}$$

The covariance matrix of the "±" variables can then be found from that of the projections as

$$B_{\pm}(\tau) = \lim_{\tau\to\infty} \langle \boldsymbol{b}_{\pm}(\tau)\boldsymbol{b}_{\pm}^{\mathrm{T}}(\tau)\rangle = \lim_{\tau\to\infty} Q^{-1}(\tau)\langle \boldsymbol{c}(\tau)\boldsymbol{c}^{\mathrm{T}}(\tau)\rangle Q^{-1\mathrm{T}}(\tau) = Q^{-1}(\tau)C(\tau)Q^{-1\mathrm{T}}(\tau). \tag{6.21}$$

From this matrix, it is now easy to move to a more physical basis, the one defined by signal and idler. From definition (6.8), we know that the relation between $\boldsymbol{b}_{\pm}$ and the fluctuations $\boldsymbol{b}$ in the signal/idler variables is $\boldsymbol{b} = \mathcal{T}\boldsymbol{b}_{\pm}$, with

$$\mathcal{T} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix}, \tag{6.22}$$

such that the covariance matrix associated to the signal/idler variables is written as

$$B(\tau) = \lim_{\tau\to\infty} \langle \boldsymbol{b}(\tau)\boldsymbol{b}^{\mathrm{T}}(\tau)\rangle = \mathcal{T}B_{\pm}(\tau)\mathcal{T}^{\mathrm{T}} = \mathcal{T}Q^{-1}(\tau)C(\tau)Q^{-1\mathrm{T}}(\tau)\mathcal{T}^{\mathrm{T}}. \tag{6.23}$$

Finally, we can write the covariance matrix associated to the quadratures of signal and idler, which is the usual object studied in the context of Gaussian states and quantum information with continuous variables. For this it is important to remind ourselves first of some relations. The quadratures are related to the annihilation and creation operators by

$$\hat{x}_j = \hat{a}_j^{\dagger} + \hat{a}_j, \qquad \hat{p}_j = \mathrm{i}(\hat{a}_j^{\dagger} - \hat{a}_j), \tag{6.24}$$

which can be written in matrix form as

$$\begin{pmatrix} \hat{x}_s \\ \hat{p}_s \\ \hat{x}_i \\ \hat{p}_i \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 & 0 & 0 \\ -\mathrm{i} & \mathrm{i} & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -\mathrm{i} & \mathrm{i} \end{pmatrix}}_{\mathcal{U}} \begin{pmatrix} \hat{a}_s \\ \hat{a}_s^{\dagger} \\ \hat{a}_i \\ \hat{a}_i^{\dagger} \end{pmatrix}. \tag{6.25}$$

Defining the fluctuations $\delta\hat{x}_j = \hat{x}_j - \langle\hat{x}_j\rangle$ and $\delta\hat{p}_j = \hat{p}_j - \langle\hat{p}_j\rangle$, which we can collect in the vector $\delta\hat{\boldsymbol{r}} = (\delta\hat{x}_s, \delta\hat{p}_s, \delta\hat{x}_i, \delta\hat{p}_i)$, the elements of the covariance matrix V are usually defined as [26]

$$V_{mn} = \frac{1}{2}\langle \delta\hat{r}_m\delta\hat{r}_n + \delta\hat{r}_n\delta\hat{r}_m\rangle, \tag{6.26}$$

It is important to write the covariance matrix as a function of normally-ordered moments, because these are the ones provided by the positive P representation, specifically through the relations

$$\langle \hat{a}_s^{\dagger m_s}\hat{a}_i^{\dagger m_i}\hat{a}_s^{n_s}\hat{a}_i^{n_i}\rangle = \langle \beta_s^{+m_s}\beta_i^{+m_i}\beta_s^{n_s}\beta_i^{n_i}\rangle / g^{(m_s+m_i+n_s+n_i)/2}. \tag{6.27}$$

However, note that the only terms out of normal in the covariance matrix appear in the diagonal, and are of the simple type $\hat{a}_j\hat{a}_j^{\dagger} = 1 + \hat{a}_j^{\dagger}\hat{a}_j = 1 + :\hat{a}_j\hat{a}_j^{\dagger}:$, where the notation ": :"

means that annihilation operators are moved to the right of creation operators (normal order). With these considerations, the asymptotic covariance matrix can be rewritten as

$$\begin{aligned}V(\tau) &= \mathcal{I} + \lim_{\tau\to\infty}\langle : \delta\hat{\boldsymbol{r}}(\tau)\delta\hat{\boldsymbol{r}}^{\mathrm{T}}(\tau) : \rangle = \mathcal{I} + \frac{1}{g}\lim_{\tau\to\infty}\mathcal{U}\langle \boldsymbol{b}(\tau)\boldsymbol{b}^{\mathrm{T}}(\tau)\rangle\mathcal{U}^{\mathrm{T}} \\ &= \mathcal{I} + \frac{1}{g}\,\mathcal{U}\mathcal{T}Q^{-1}(\tau)C(\tau)Q^{-1\mathrm{T}}(\tau)\mathcal{T}^{\mathrm{T}}\mathcal{U}^{\mathrm{T}},\end{aligned} \tag{6.28}$$

where we have also made use of the relation (6.25) between quadratures and bosonic operators. This is the final expression that we will use in order to analyze the quantum properties of the system, starting by the quantum steady state in the next section.

Finally, it is worth noting that the covariance matrix found above is independent of g (since C is proportional to g). This is typical of the predictions coming out of linearized theories, and shows that such a level of approximation must necessarily break down when quantum noise becomes too large, that is, when g is not small.

6.6 Quantum steady state and phase-space distribution

With the results of the previous section at hand, now we can build the approximation to the quantum steady state spilled out by the method, which we already introduced qualitatively in (6.2). We found in the previous section that our linearization provides a Gaussian state $\hat{\rho}_G(\tau)$ that evolves periodically in time, with covariance matrix (6.28) and mean vector $\boldsymbol{d} = \langle(\hat{x}_s, \hat{p}_s, \hat{x}_i, \hat{p}_i)^{\mathrm{T}}\rangle$ given by

$$\boldsymbol{d}(\tau) = 2\big(\mathrm{Re}\{\bar{\beta}_s(\tau)\}, \mathrm{Im}\{\bar{\beta}_s(\tau)\}, \mathrm{Re}\{\bar{\beta}_i(\tau)\}, \mathrm{Im}\{\bar{\beta}_i(\tau)\}\big)^{\mathrm{T}}, \tag{6.29}$$

which simply follows the trajectory of the classical limit cycle. As argued in the introduction, and shown explicitly in the previous section, for long times it is impossible to predict where the system is along this trajectory, since θ is completely undetermined (variance tends to infinity). In other words, asymptotically, at a given time we cannot predict which of the Gaussian states $\hat{\rho}_G(\tau)$ describes the system, so that the true quantum mechanical state of the system is a balanced mixture of all of them, as given in (6.2), that is

$$\bar{\rho} = \lim_{\tau\to\infty}\hat{\rho}(\tau) = \int_0^T \frac{d\tau}{T}\hat{\rho}_G(\tau). \tag{6.30}$$

A convenient way to write this state is through its Wigner function [11, 26], which also allows us to visualize and interpret it in phase space. The Wigner function of the Gaussian states $\hat{\rho}_G(\tau)$ can be written as

$$W_G(\boldsymbol{r},\tau) = \frac{1}{2\pi\sqrt{\det\{V(\tau)\}}}e^{-\frac{1}{2}\left(\boldsymbol{r}-\boldsymbol{d}(\tau)\right)^{\mathrm{T}}V^{-1}(\tau)\left(\boldsymbol{r}-\boldsymbol{d}(\tau)\right)}, \tag{6.31}$$

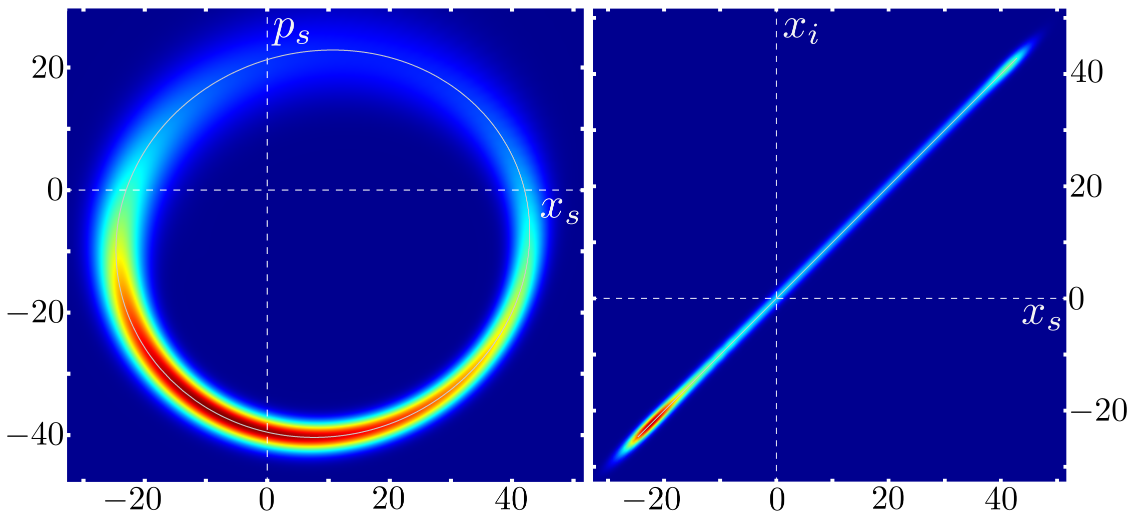

FIGURE 6.2. Marginals of the Wigner distribution corresponding to the quantum steady state provided by our linearized theory. We chose the same parameters as in Fig. 6.1, and $g = 0.00025$.

where $\boldsymbol{r} = (x_s, p_s, x_i, p_i)$ is the coordinate vector in phase space. The Wigner function of the mixture (6.30) can then be written as

$$\bar{W}(\boldsymbol{r}) = \int_0^T \frac{d\tau}{T} W_G(r, \tau). \tag{6.32}$$

The first thing to note is that, being a mixture of Gaussians, this distribution is positive and bounded, and therefore, it is a true probability distribution in phase space. Specifically, it is the probability density function describing the statistics of quadrature measurements. On the other hand, since this is a function in a four-dimensional space, we visualize it in Fig.6.2 through two-dimensional marginals of the distribution, by integrating two of the phase-space variables. Two things are worth noting. First, note that the distribution is localized around the classical limit cycle, as clearly appreciated in the left panel, where the classical trajectory is shown in white. Second, the distribution shows correlations between the position quadratures of signal and idler (see the right panel), and a similar thing is found for the momenta, which are anticorrelated. In the next chapter we study these correlations in detail, showing that they correspond to quantum correlations (entanglement).

CHAPTER 7

QUANTUM CORRELATIONS AND OBSERVATIONAL CONSIDERATIONS

In this chapter we analyze detail the Gaussian states that the linearization method allowed us to develop at each point of the cycle's trajectory. We start by providing a quick introduction to quantum correlations and entanglement, presenting a criterion capable of evidencing it in Gaussian states. We then apply the criterion to our Gaussian states, and prove that they contain entanglement levels close to the ideal ones attainable with standard stationary non-degenerate OPOs. Finally we discuss some subtle issues of the Gaussian states, specifically their apparent violation of the uncertainty principle. We provide an explanation for this, and also discuss its implications for the observation and use of the quantum correlations that we put forward in the Gaussian states.

7.1 EPR, two-mode squeezing, and entanglement criterion

One of the most mysterious and interesting predictions of quantum mechanics is entanglement. In the introductory chapter we discussed some of the conclusions drawn in the EPR paper [2]. The authors introduced a state aimed at proving that quantum mechanics provides an incomplete description of reality. Nowadays, we know that the arguments put forward by the EPR paper are flawed, but indeed the state that they constructed has remained as one of the cornerstones of quantum mechanics, since it provided the first clear example of an entangled state. Let us then start introducing the EPR state, which is defined for two particles moving in one dimension, e.g., two harmonic oscillators or, equivalently, two modes of the electromagnetic field such as our signal and idler. Denoting by $\{|x\rangle\}_{x\in\mathbb{R}}$ and $\{|p\rangle\}_{p\in\mathbb{R}}$, respectively, the eigenstates of the position and momentum quadratures defined in (6.24), EPR defines the pure state

$$|\mathrm{EPR}\rangle = \int_{\mathbb{R}} dx|x\rangle \otimes |x\rangle = \int_{\mathbb{R}} dp|p\rangle \otimes |-p\rangle, \tag{7.1}$$

where in the second equality we have used the following well-known relation between the position and momentum eigenstates

$$|x\rangle = \int_{\mathbb{R}} dp \frac{\exp(\mathrm{i}xp/2)}{\sqrt{4\pi}}|p\rangle. \tag{7.2}$$

There are several things to note about this state. First, it is not normalizable, and therefore, rigorously speaking it is not physical. However, we will show later that it can be extremely well approximated by a physical state (two-mode squeezed vacuum), which converges to it in the unphysical infinite-energy limit. Second, upon a measurement of the position of one of the modes, all possible values are equally likely, that is, the probability density function for single-mode position measurements is a uniform distribution. However, once we measure a specific value x_0 for the first mode, we know with certainty that a measurement of the position of the second oscillator will provide x_0 as well. The same applies to the momentum, all values are equaly likely when measuring the momentum of the first mode, but once we measure p_0, we know with certainty that the second mode will have momentum $-p_0$. Therefore, this state provides a maximum degree of correlation between noncommuting observables of the two modes. This is a type of correlation that cannot be obtained without using the superposition principle, and hence it goes beyond the type of correlations that are allowed in classical mechanics. We therefore refer to them as quantum correlations or *entanglement*.

Let us now explain one way of generating physical states that approximate the EPR state. For this, we use the two mode squeezing operator, defined as

$$\hat{S}(r) = e^{r(\hat{a}_s^\dagger \hat{a}_i^\dagger - \hat{a}_s\hat{a}_i)}, \tag{7.3}$$

where $r \in \mathbb{R}$ is known as the squeezing parameter, taken real here for simplicity. This is a unitary operator that corresponds to letting evolve the system according to the non-degenerate down-conversion Hamiltonian (2.5) during a time $t = r/G$ (taking G real), and therefore, we expect the conclusions drawn from the action of this operator to apply in our system to a certain degree. In order to show the connection between this operator and the EPR state, let us consider the evolution in Heisenberg picture, where the annihilation operators transform as [26]

$$\hat{a}_s(r) = \hat{S}^\dagger(r)\hat{a}_s(0)\hat{S}(r) = \hat{a}_s(0)\cosh r + \hat{a}_i^\dagger(0)\sinh r, \tag{7.4a}$$

$$\hat{a}_i(r) = \hat{S}^\dagger(r)\hat{a}_i(0)\hat{S}(r) = \hat{a}_i(0)\cosh r + \hat{a}_s^\dagger(0)\sinh r, \tag{7.4b}$$

leading to the quadratures

$$\hat{x}_s(r) = \hat{x}_s(0)\cosh r + \hat{x}_i(0)\sinh r, \tag{7.5a}$$
$$\hat{x}_i(r) = \hat{x}_i(0)\cosh r + \hat{x}_s(0)\sinh r, \tag{7.5b}$$
$$\hat{p}_s(r) = \hat{p}_s(0)\cosh r - \hat{p}_i(0)\sinh r, \tag{7.5c}$$
$$\hat{p}_i(r) = \hat{p}_i(0)\cosh r - \hat{p}_s(0)\sinh r. \tag{7.5d}$$

Then, we find that the difference of positions and the sum of momenta evolve as

$$\hat{x}_s(r) - \hat{x}_i(r) = (\hat{x}_s(0) - \hat{x}_i(0))e^{-r}, \tag{7.6a}$$
$$\hat{p}_s(r) + \hat{p}_i(r) = (\hat{p}_s(0) + \hat{p}_i(0))e^{-r}, \tag{7.6b}$$

from which we obtain the variances

$$V\big(\hat{x}_s(r) - \hat{x}_i(r)\big) = V\big(\hat{x}_s(0) - \hat{x}_i(0)\big)e^{-2r}, \tag{7.7a}$$
$$V\big(\hat{p}_s(r) + \hat{p}_i(r)\big) = V\big(\hat{p}_s(0) + \hat{p}_i(0)\big)e^{-2r}. \tag{7.7b}$$

We observe that, irrespectively of the initial state, as the squeezing parameter tends to infinity the positions (momenta) of the modes get perfectly correlated (anticorrelated), just as in the EPR state. Indeed, it is possible to show that starting from vacuum, we obtain exactly the $|\mathrm{EPR}\rangle$ state in the $r \to \infty$ limit [26]. However, this limit is unphysical, as evidenced by the fact that the number of excitations in either of the modes increases as $\sinh^2 r$ [26], meaning that we would require infinite energy to create it. In any case, even if not perfect, quantum correlations can still be very high for physical states. Indeed, we will show in the next section that our OPO shows EPR entanglement between signal and idler.

It is finally interesting to note that Duan, Giedke, Cirac, and Zoller [27] and Simon [28] derived an entanglement criterion which is a sufficient and necessary condition for two-mode Gaussian states, and can be seen as a generalization of the EPR correlations. This criterion states that the Gaussian state is entangled if and only if there exist parameters $\phi_s \in [0, 2\pi]$, $\phi_i \in [0, 2\pi]$, and $k \in \mathbb{R} \neq 0$ such that

$$V\left(\frac{k\hat{x}_s^{\phi_s} - k^{-1}\hat{x}_i^{\phi_i}}{\sqrt{2}}\right) + V\left(\frac{k\hat{p}_s^{\phi_s} + k^{-1}\hat{p}_i^{\phi_i}}{\sqrt{2}}\right) < k^2 + k^{-2} \tag{7.8}$$

where

$$\hat{x}_j^{\phi} = e^{i\phi}\hat{a}_j^\dagger + e^{-i\phi}\hat{a}_j = \hat{x}_j\cos\phi + \hat{p}_j\sin\phi, \tag{7.9}$$
$$\hat{p}_j^{\phi} = \mathrm{i}(e^{\mathrm{i}\phi}\hat{a}_j^\dagger - e^{-\mathrm{i}\phi}\hat{a}_j) = \hat{p}_j\cos\phi - \hat{x}_j\sin\phi, \tag{7.10}$$

are just the position and momentum quadratures in a rotated phase-space frame.

In the following we discuss the properties of the Gaussian states $\hat{\rho}_G(\tau)$ that we build in the previous chapter from the point of view of this criterion.

7.2 Quantum correlations of our Gaussian states

From the analysis of [9], we know that there are large levels of entanglement at the Hopf bifurcation of our system. On the other hand, in the absence of injection ($\mathcal{P}=0$), we also know that quantum correlations are large, since we recover the standard non-degenerate OPO [18]. Therefore, we expect to find large levels of quantum correlations for the limit cycles that connect these two points, corresponding to the top-right and bottom-left panels of Fig. 5.1. We therefore stay within the corresponding parameters.

All the properties of our Gaussian states $\hat{\rho}_G(\tau)$ are encoded in the covariance matrix $V(\tau)$ given by (6.28), which we can evaluate efficiently at all points of the cycle's trajectory using the Floquet method as we explained in the previous chapter. On the other hand, we have checked numerically that for any value of the model parameters (within the regions specified above), the eigenvectors of $V(\tau)$ can be written as

$$\boldsymbol{v}_0(\tau)=\frac{1}{\sqrt{2}}\Big(\sin\varphi(\tau),\cos\varphi(\tau),\sin\varphi(\tau),-\cos\varphi(\tau)\Big)^{\mathrm{T}}, \tag{7.11a}$$

$$\boldsymbol{v}_1(\tau)=\frac{1}{\sqrt{2}}\Big(\cos\varphi(\tau),-\sin\varphi(\tau),\cos\varphi(\tau),\sin\varphi(\tau)\Big)^{\mathrm{T}}, \tag{7.11b}$$

$$\boldsymbol{v}_2(\tau)=\frac{1}{\sqrt{2}}\Big(\sin\phi(\tau),\cos\phi(\tau),-\sin\phi(\tau),\cos\phi(\tau)\Big)^{\mathrm{T}}, \tag{7.11c}$$

$$\boldsymbol{v}_3(\tau)=\frac{1}{\sqrt{2}}\Big(\cos\phi(\tau),-\sin\phi(\tau),-\cos\phi(\tau),-\sin\phi(\tau)\Big)^{\mathrm{T}}. \tag{7.11d}$$

where $\phi(\tau)$ and $\varphi(\tau)$ depend on the choice of system parameters, but are easily found numerically for any choice. Defining the vector operator $\hat{\boldsymbol{r}}=(\hat{x}_s,\hat{p}_s,\hat{x}_i,\hat{p}_i)^{\mathrm{T}}$, these eigenvectors allow us to define the following combinations of quadratures:

$$\hat{X}_0(\tau)=\boldsymbol{v}_0^{\mathrm{T}}(\tau)\hat{\boldsymbol{r}}=\frac{1}{\sqrt{2}}(\hat{p}_s^{-\varphi(\tau)}-\hat{p}_i^{\varphi(\tau)}), \tag{7.12a}$$

$$\hat{X}_1(\tau)=\boldsymbol{v}_1^{\mathrm{T}}(\tau)\hat{\boldsymbol{r}}=\frac{1}{\sqrt{2}}(\hat{x}_s^{-\varphi(\tau)}+\hat{x}_i^{\varphi(\tau)}), \tag{7.12b}$$

$$\hat{X}_2(\tau)=\boldsymbol{v}_2^{\mathrm{T}}(\tau)\hat{\boldsymbol{r}}=\frac{1}{\sqrt{2}}(\hat{p}_s^{-\phi(\tau)}+\hat{p}_i^{\phi(\tau)}), \tag{7.12c}$$

$$\hat{X}_3(\tau)=\boldsymbol{v}_3^{\mathrm{T}}(\tau)\hat{\boldsymbol{r}}=\frac{1}{\sqrt{2}}(\hat{x}_s^{-\phi(\tau)}-\hat{x}_i^{\phi(\tau)}). \tag{7.12d}$$

The variance of these operators is given by the corresponding eigenvalues, which we denote by $\{V_n(\tau)\}_{n=0,1,2,3}$. This is easily proven by computing the elements of the covariance matrix of the combined quadratures as follows

$$\frac{\langle\delta\hat{X}_j\delta\hat{X}_l+\delta\hat{X}_l\delta\hat{X}_j\rangle}{2}=\sum_{m,n=1}^{4}(\boldsymbol{v}_j)_m(\boldsymbol{v}_l)_n\frac{\langle\delta\hat{r}_m\delta\hat{r}_n+\delta\hat{r}_n\delta\hat{r}_m\rangle}{2}=\boldsymbol{v}_j^{\mathrm{T}}V\boldsymbol{v}_l=V_j\delta_{jl}, \tag{7.13}$$

where we have made use of (7.12), the eigenvector relation $V\boldsymbol{v}_j=V_j\boldsymbol{v}_j$, and the orthonormality relations $\boldsymbol{v}_j^{\mathrm{T}}\boldsymbol{v}_l=\delta_{jl}$ (note that the covariance matrix is real and symmetric). We plot the eigenvalues in Fig. 7.1. In general, for any choice of the parameters we obtain

$$V_0(\tau) = 1, \quad V_1(\tau) > 1, \quad V_2(\tau) < 1, \quad V_3(\tau) < 1. \tag{7.14}$$

This provides us with a very interesting structure for the correlations of the Gaussian states along the cycle's trajectory. First, note that there is one quadrature combination, $\hat{X}_0$, that has the variance of vacuum at all points of the trajectory. This is a clear consequence of having set the projection $c_0(\tau) = 0$ in the previous chapter, and later on we will comment more on it. On the other hand, we have two quadrature combinations $\hat{X}_2$ and $\hat{X}_3$, which are squeezed below the vacuum noise level (variance below 1) at all points of the trajectory. In addition, these quadratures form a pair that matches exactly the pair appearing in the entanglement criterion (7.8) with $k = 1$ and $\phi_i = -\phi_s = \phi(\tau)$. Therefore, the criterion states that our Gaussian states are indeed entangled.

In order to get an idea of how much entanglement do we obtain, let us compare with the values that can be found in stationary non-degenerate OPOs above threshold [9, 18]. Let's consider for simplicity the ideal situation with $\tilde{\Delta} = 0$ (no detuning) and $\mathcal{P} = 0$ (no subharmonic injection). In such case the classical solution reads $\bar{\beta}_s = \bar{\beta}_i = \sqrt{\sigma - 1}$ (up to an arbitrary relative phase), and one finds the same eigenvectors (7.11) for the covariance matrix, but with $\phi(\tau) = 0 = \varphi(\tau)$. The corresponding variances are easily found to be $V_3 = 0.5$ and $V_2 = 1 - (1/2\sigma)$. Interestingly, both converge to 0.5 at threshold $\sigma = 1$, which translates to perfect noise reduction when analyzing the squeezing spectrum outside the cavity [18]. The fact that V_3 is fixed to that ideal 0.5 value comes from the perfect intensity correlations that exist between signal and idler: for $\mathcal{P} = 0$, all the signal/idler light comes from down-conversion, which generates signal/idler photons in pairs. This is the same even in the presence of detuning, but not in the presence of injection ($\mathcal{P} \neq 0$), which introduces coherent uncorrelated signal and idler light, and therefore degrades their correlations. Still, in [9] it was shown that the squeezing degradation is kept small at the Hopf bifurcation as long as the detuning is not excessively large. V_2, on the other hand, depends on the distance to threshold, and in particular squeezing is lost as σ becomes larger, reaching the level of vacuum fluctuations ($V_2 = 1$) asymptotically. Hence, in order to get large levels of entanglement, one cannot go arbitrarily far away from threshold. Nevertheless, it was found in [9] that even in the presence of injection and detuning, at the Hopf bifurcation this quadrature $\hat{X}_2$ shows similar squeezing levels as $\hat{X}_3$. Hence, close-to-ideal entanglement is still present at the Hopf bifurcation of the actively-phase-locked OPO, so we should expect this to still be true to certain extent in the limit-cycle region that we are studying. Indeed, it can be appreciated in the bottom panel of Fig. 7.1 that $V_2(\tau)$ and $V_3(\tau)$ remain close to their ideal 0.5 value, with $V_3 < V_2$ as expected from the previous discussion. In summary, we see that the Gaussian states retain the entanglement levels typical from stationary configurations of non-degenerate OPOs.

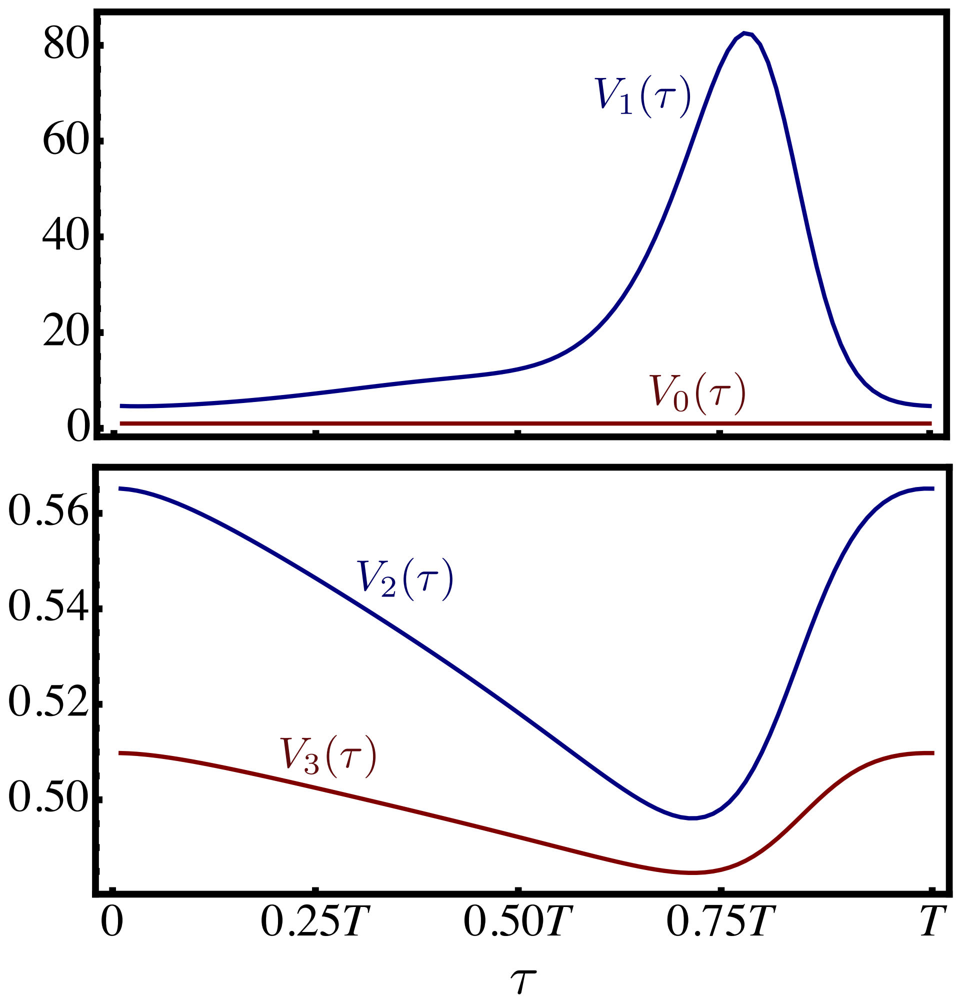

FIGURE 7.1. Variance of the eigen-quadratures of the covariance matrix $V(\tau)$ of the Gaussian states defined at each point of the cycle's trajectory (we choose the same system parameters as in the previous figures). In the upper panel we show $V_0(\tau)$ in red and $V_1(\tau)$ in blue. The lower panel shows $V_3(\tau)$ in red and $V_2(\tau)$ in blue. As mentioned in the text, at all points of the trajectory we observe that $V_0(\tau) = 1$, $V_1(\tau) > 1$, while $V_2(\tau)$ and $V_3(\tau)$ are close to 0.5. The latter means that our Gaussian states are entangled, since the combined quadratures $\hat{X}_2(\tau)$ and $\hat{X}_3(\tau)$ have the form required by the entanglement criterion (7.8).

7.3 Physical and observational considerations

In the discussion provided above, we have omitted a big issue: the Gaussian states that the method provides seem to violate the uncertainty principle, and hence, cannot be rendered as physical states. In order to see this, note that the following commutators between the combined quadratures (7.12) are non-zero:

$$[\hat{X}_3(\tau), \hat{X}_0(\tau)] = [\hat{X}_1(\tau), \hat{X}_2(\tau)] = 2\mathrm{i}\cos\big(\phi(\tau) - \varphi(\tau)\big), \tag{7.15a}$$

$$[\hat{X}_2(\tau), \hat{X}_0(\tau)] = [\hat{X}_3(\tau), \hat{X}_1(\tau)] = 2\mathrm{i}\sin\big(\phi(\tau) - \varphi(\tau)\big). \tag{7.15b}$$

On the other hand, for all the parameters that we have analyzed, and as appreciated from Fig. 7.2, we have found that our method provides variances that satisfy the following properties:

$$V_3(\tau)V_0(\tau) \leq \cos^2\big(\phi(\tau) - \varphi(\tau)\big), \tag{7.16a}$$

$$V_1(\tau)V_2(\tau) \geq \cos^2\big(\phi(\tau) - \varphi(\tau)\big), \tag{7.16b}$$

$$V_2(\tau)V_0(\tau) \geq \sin^2\big(\phi(\tau) - \varphi(\tau)\big), \tag{7.16c}$$

$$V_3(\tau)V_1(\tau) \geq \sin^2\big(\phi(\tau) - \varphi(\tau)\big). \tag{7.16d}$$

Taking into account that the uncertainty principle reads $V(\hat{A})V(\hat{B}) \geq |\langle[\hat{A}, \hat{B}]\rangle|^2/4$, the last three properties agree with it. However, the first property (7.16a) seems to violate it. Therefore we seem to be forced to conclude that the linearization method provides non-physical Gaussian states, which puts the whole analysis at jeopardy. Fortunately, the story is far more subtle than this, and next we elaborate on it.

The solution to this conundrum can be traced back to our expansion (6.3) when defining the quantum fluctuations. Specifically, in the time argument of the fluctuations we are including the stochastic variable $\theta(\tau)$, which diffuses with time, as proved in the previous chapter. In other words, the linearized equations (6.4) are implicitly written in a frame co-moving with the random diffusion occurring along the cycle. Therefore, the quadratures defined in (7.12) are also defined in this co-moving frame, where the commutation relations (7.15) are not the true ones. This is indeed exactly the same situation that was pointed out in [24, 25], but there in the context of a different spontaneously broken symmetry, rotations in space instead of translations in time. In that case, it was possible to show analytically that the '$\theta(\tau)$-co-moving' Gaussian states $\rho_G(\tau)$ indeed satisfy the uncertainty relations corresponding to the co-moving frame. The same is expected to be true in our current system, but given that our symmetry is far more complicated, this is something that we are in no position to prove at this time.

This opens up questions regarding the observability of the entanglement that we have predicted. Again, this is a question that it's too difficult to rigorously tackle in our system, but thankfully it was studied in depth in the context of spontaneous rotational symmetry breaking. The same conclusions are expected to apply to our case, which we summarize next. Technically speaking, our predictions for quantum correlations apply only in the frame co-moving with $\theta(\tau)$, which is a random variable and therefore no experiment can really make observations in that reference frame. However, it was proven in [25] that as long as $\theta(\tau)$ is initially locked to a known value (which later we explain how to do) and its

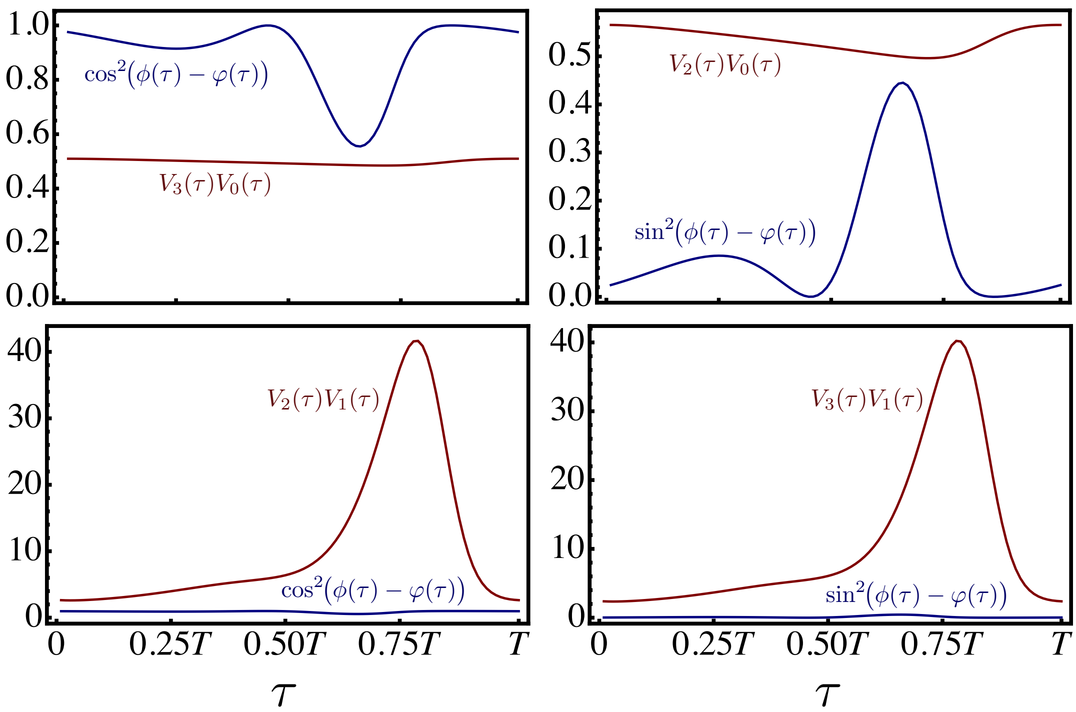

FIGURE 7.2. Checking the uncertainty relations of our Gaussian states. On each plot we show the product of variances of two combined quadratures $V_m(\tau)V_n(\tau)$ (red) and the corresponding value for the expected right-hand side of the uncertainty principle, $\langle[\hat{X}_m(\tau),\hat{X}_n(\tau)]\rangle|^2/4$ (blue), which is equal to $\cos^2\left(\phi(\tau)-\varphi(\tau)\right)$ in the left panels and $\sin^2\left(\phi(\tau)-\varphi(\tau)\right)$ in the right pannels. Specific values of (m,n) are as follows: $(3,0)$ in the top-left panel, $(2,0)$ in the top-right panel, $(2,1)$ in the bottom-left panel, and $(3,1)$ in the bottom-right panel. We can appreciate that the Gaussian states seem to incur in a violation of the uncertainty principle, as evidenced by the results of the top-left panel, rendering these states unphysical. In the main text we explain why this is not exactly the case.

diffusion is slow (which is usually the case in OPO experiments, as g is very small), large levels of quantum correlations can still be measured in the laboratory frame. Moreover, θ can be locked to a fixed value by adding a term in the equations of motion which explicitly breaks their symmetry, which in our case can be done by adding a sideband to any of the lasers, which would induce a time-dependent $\epsilon_j(\tau)$ in the stochastic equations (4.42). Of course, this comes at the price of reducing the quantum correlations in the system, but in the regime where the symmetry-breaking term acts as a small perturbation as compared with g^{-1}, large levels of quantum correlations are still expected to appear. This was analytically proven in [29, 30] for spatial rotational symmetry.

In summary, the predictions obtained through the Gaussian states that we built in the reference frame co-moving with $\theta(\tau)$ are expected to apply with just small modifications in the laboratory frame.

Chapter 8

Conclusions and outlook

In this thesis we made the first steps in the development of the quantum theory of actively-phase-locked type II optical parametric oscillators, under conditions where they undergo limit-cycle motion.

We develop the quantum master equation that rules their dynamics, as well as the corresponding dynamical equation for their phase-space quasiprobability distributions. We showed that using the positive P representation, the master equation can be exactly mapped to a set of stochastic Langevin equations. Even though these are still not analytically solvable, they allowed us to perform an approximate linearized analysis expected to work well when quantum noise acts as a small perturbation, as is usually the case in OPOs.

Following [9], we first analyzed the system in the classical limit, finding the region of parameters where the OPO undergoes limit-cycle motion. We then proceeded to apply the linearized theory of quantum fluctuations. However, in its standard form developed around classical stationary solutions, it is not suitable for our problem, since quantum fluctuations along the cycle's phase-space trajectory cannot be considered small. Nevertheless, a linearization technique around limit cycles was developed in [10], which we applied here for the first time in an experimentally relevant scenario, since actively-phase-locked optical parametric oscillators are very mature nonlinear devices. In essence, the technique is capable of tracking the variable accounting for the nonlinear fluctuations along the trajectory, which is shown to diffuse more and more as time goes by. By moving to a reference frame co-moving with this diffusion, one can then make a linear expansion in the quantum fluctuations, which in combination with Floquet theory, provides a Gaussian state at each point of the cycle's trajectory. When coming back to the laboratory reference frame, the quantum steady state of the master equation was then approximated by the balanced mixture of these Gaussian states.

Finally, we analyzed the quantum fluctuations present in the Gaussian states, showing that they present nontrivial quantum correlations in the form of EPR entanglement. However, these states are formulated in the frame co-moving with the diffusion along the cycle, which is not accessible in experiments. Based on the studies performed in a similar scenario, we then argued that large levels of entanglement should still be present in the laboratory frame [24, 25], or even by stopping the diffusion by adding a sideband to the lasers, which would explicitly break the symmetry of the equations under time translations [29, 30].

Since our analysis has focussed on the intracavity field, the obvious next step will be the analysis of the light coming out from the cavity, and specific experimentally-relevant observable quantities such as the squeezing spectra and the related spectral covariance matrix [9]. This will require the computation of dynamical quantities such as two-time correlation functions both in the laboratory frame and the frame co-moving with the diffusion, which are still open problems in this scenario.

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